

Lecture 9: Quantum mechanics and atomic structure 3

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Classroom: Teaching Center 401

Wednesday/Friday: 10:15-11:00; 11:10-11:55

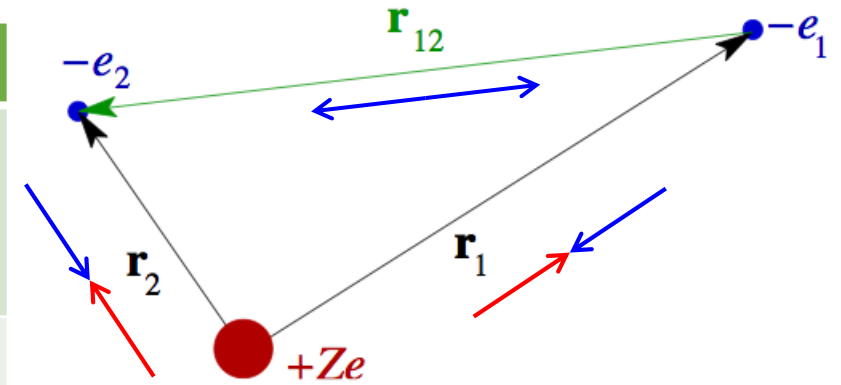
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Major points of last lecture

- **Quantum mechanics and atomic structure**
 1. The hydrogen atom
 2. Shell model for many-electron atoms
 3. Aufbau principle and electron configuration
 4. Shells and the periodic table: photoelectron spectroscopy
 5. Periodic properties and electronic structure

Shell model for many-electron atoms: Single- vs. Many-Electron Atoms

	H, He ⁺ , Li ²⁺	He, Li, ...
Interactions	Z-e attraction	Z-e attraction e-e repulsion electron spins
Orbitals	e in a single orbital	e in multiple orbitals
Coordinates	Z at O, e at r	Z at O, e_1 at r_1 , e_2 at r_2 , ...
Schrödinger Eqn.	Analytical solutions	Approximate solutions



An electron is attracted by $+Z$ but repelled by $-(Z-1)e$

$$\psi = \psi(x_1, y_1, z_1) \quad \left[-\frac{\hbar^2}{2m} \nabla^2 - \frac{Ze^2}{4\pi\epsilon_0 r} \right] \psi = E\psi$$

$$\psi = \psi(x_1, y_1, z_1, x_2, y_2, z_2) \quad \frac{-\hbar^2}{2m_e} \left[\left(\frac{\partial^2 \psi}{\partial x_1^2} + \frac{\partial^2 \psi}{\partial y_1^2} + \frac{\partial^2 \psi}{\partial z_1^2} \right) + \left(\frac{\partial^2 \psi}{\partial x_2^2} + \frac{\partial^2 \psi}{\partial y_2^2} + \frac{\partial^2 \psi}{\partial z_2^2} \right) \right] + \left(-\frac{2e^2}{4\pi\epsilon_0 r_1} - \frac{2e^2}{4\pi\epsilon_0 r_2} + \frac{e^2}{4\pi\epsilon_0 r_{12}} \right) \psi = E\psi$$

$$\psi = \psi(x_1, y_1, z_1, x_2, y_2, z_2, x_3, y_3, z_3)$$

Shell model for many-electron atoms: Hartree Orbitals

Self-consistent field (SCF) orbital approximation method (自洽场近似法):

1. generates **a set of approximate one-electron orbitals** (ψ_α), and associated energy levels (ϵ_α), reminiscent of those for the H atom.
2. The electronic structure of an atom with atomic number Z is then “built up” by placing Z electrons into these orbitals in accordance with certain rules

$$\frac{-\hbar^2}{2m_e} \left[\left(\frac{\partial^2 \psi}{\partial x_1^2} + \frac{\partial^2 \psi}{\partial y_1^2} + \frac{\partial^2 \psi}{\partial z_1^2} \right) + \left(\frac{\partial^2 \psi}{\partial x_2^2} + \frac{\partial^2 \psi}{\partial y_2^2} + \frac{\partial^2 \psi}{\partial z_2^2} \right) \right] + \left(-\frac{2e^2}{4\pi\epsilon_0 r_1} - \frac{2e^2}{4\pi\epsilon_0 r_2} + \frac{e^2}{4\pi\epsilon_0 r_{12}} \right) \psi = E\psi$$

$$\psi = \psi(x_1, y_1, z_1, x_2, y_2, z_2)$$

$$\varphi_\alpha(r_1)\varphi_\beta(r_2)$$

Shell model for many-electron atoms: Hartree Orbitals

Hartree's method: three assumptions

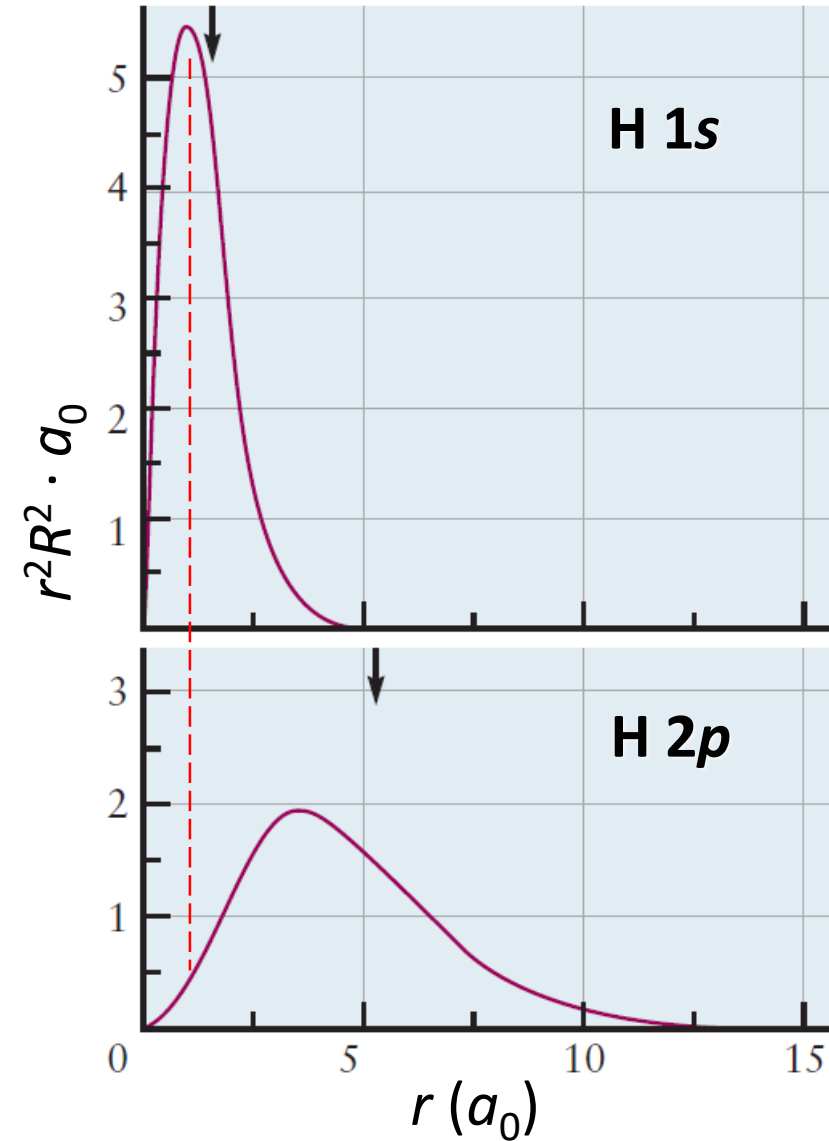
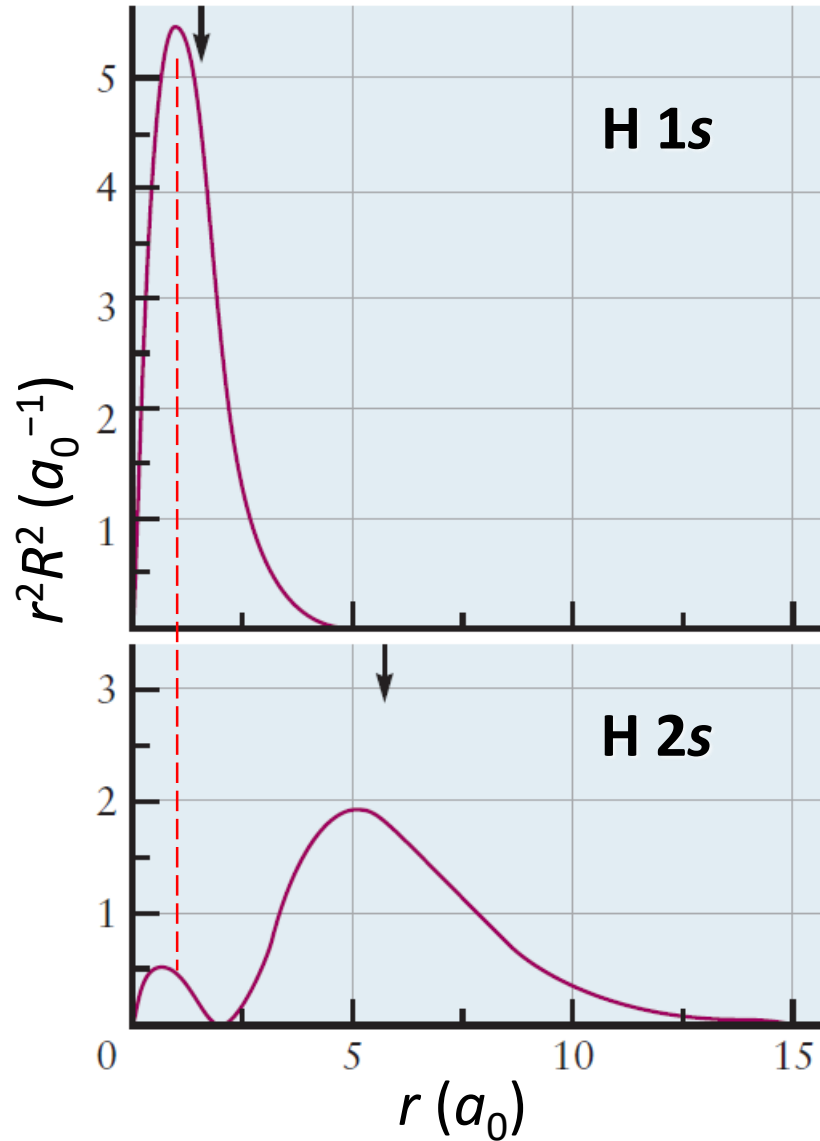
- Each electron moves in **an effective field** created by the nucleus and all the other electrons, and the effective field for electron i **depends only on its position** r_i .
- The effective field for electron i is obtained by averaging its Coulomb potential interactions with each of the other electrons over all the positions of the other electrons: **r_i is the only coordinate in the description.**
- The effective field is spherically symmetric: no angular dependence.

Li atom $\psi = \psi(x_1, y_1, z_1, x_2, y_2, z_2, x_3, y_3, z_3)$

The orbital approximation for atoms $\psi_{\text{atom}} = \varphi_{\alpha}(r_1)\varphi_{\beta}(r_2)\varphi_{\gamma}(r_3)$

The 2nd and 3rd assumptions in effect convert the exact Schrödinger equation for the atom into a set of simultaneous equations for the **unknown effective field and the unknown one-electron orbitals**. These equations must be **solved by iteration until a self-consistent solution** is obtained

The Penetration Effect 钻穿效应



Shielding Effects: Energy Sequence of Hartree Orbitals

The dependence of the energy on ℓ and n can be explained by comparing the extent of shielding in different subshells.

$$\epsilon_{ns} < \epsilon_{np} < \epsilon_{nd}$$

- Only the **s orbitals penetrate** to the nucleus
- Both **p and d orbitals have nodes at the nucleus**
- Consequently, the shielding will be smallest, and the electron most tightly bound, in s orbitals

Shielding Effects: Energy Sequence of Hartree Orbitals

Assume each of the electrons in shell n is moving in a Coulomb potential given approximately by

$$V_n^{\text{eff}}(r) \approx -\frac{Z_{\text{eff}}(n)e^2}{r}$$

We use ε_n to distinguish a Hartree orbital energy from the H atom orbital E_n

$$\varepsilon_n \approx -\frac{[Z_{\text{eff}}(n)]^2}{n^2} \quad (\text{rydbergs})$$

$$\bar{r}_{n\ell} \approx \frac{n^2 a_0}{Z_{\text{eff}}(n)} \left\{ 1 + \frac{1}{2} \left[1 - \frac{\ell(\ell + 1)}{n^2} \right] \right\}$$

Shielding Effects: Energy Sequence of Hartree Orbitals

Slater's **empirical rules** for **estimating the shielding constant** for an electron in an ns or np orbital:

1. Write the electronic configuration for the atom with terms **in the order and grouped**: (1s) (2s, 2p) (3s, 3p) (3d) (4s, 4p) (4d) (4f) (5s, 5p) etc.
2. **Electrons in any group to the right of a (ns, np) group do not shield the (ns, np) electrons** and therefore make no contribution to the shielding constant.
3. **Each of the other electrons in a (ns, np) group contributes 0.35** to the shielding constant for the valence electron.
4. **Each electron in the (n-1) shell contributes 0.85** to the shielding constant.
5. **Each electron in shells (n-2) or lower fully shield the outer electrons**, contributing 1.00 to the shielding constant.

Shielding Effects: Energy Sequence of Hartree Orbitals

Orbital Energies (×13.6 eV)

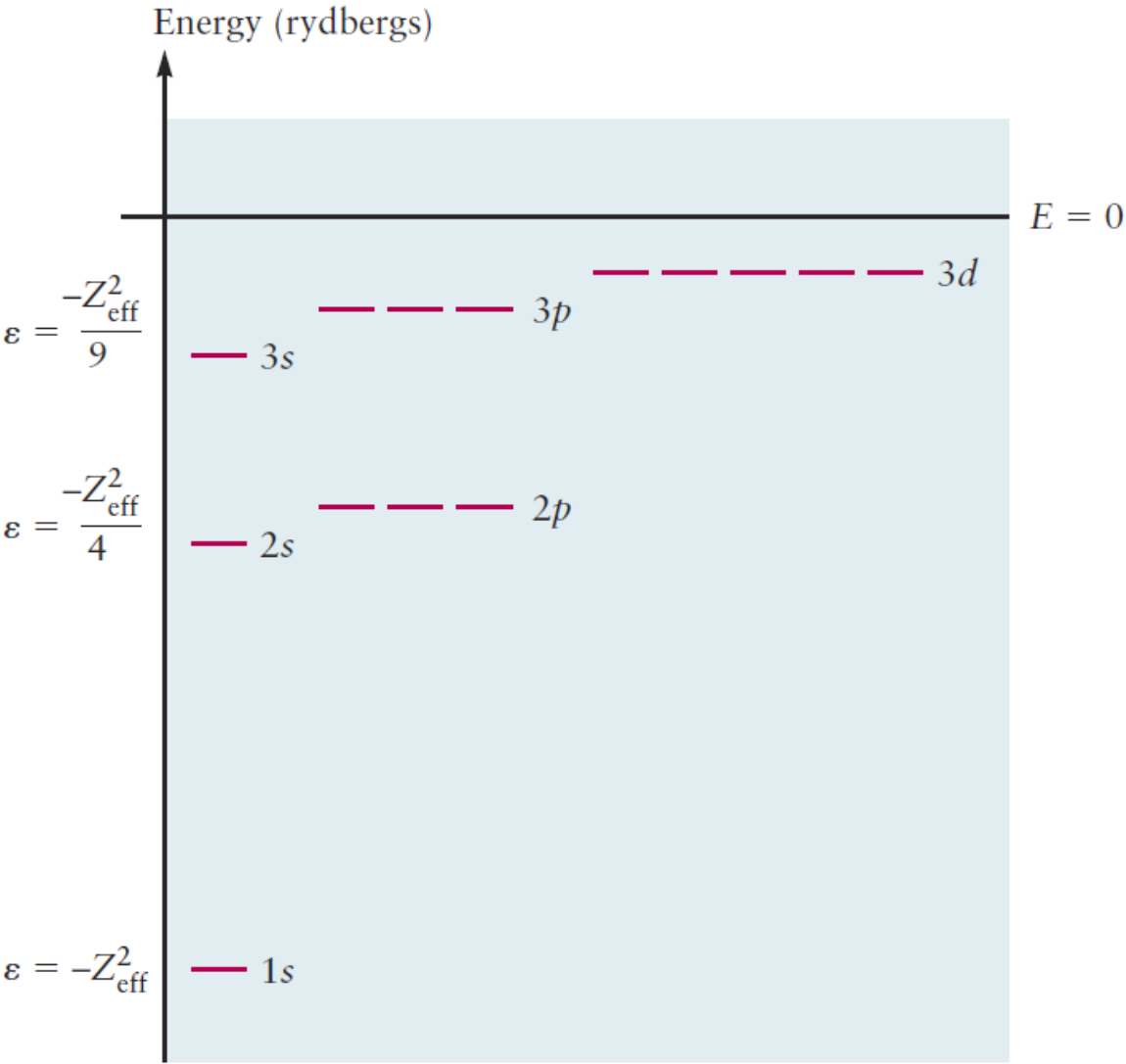
	¹ H	² He ⁺	³ Li ²⁺	⁴ Be ³⁺	⁵ B ⁴⁺
1s	−1	−2 ²	−3 ²	−4 ²	−5 ²
2s, 2p	−1/2 ²	−(2/2) ²	−(3/2) ²	−(4/2) ²	−(5/2) ²

	H	He	Li	Be	B
1s	−1	−1.65 ²	−2.65 ²	−3.65 ²	−4.65 ²
2s			−(1.3/2) ²	−(1.95/2) ²	−(2.6/2) ²
2p					−(2.6/2) ²

TABLE 5.3
Z_{eff} for Selected Atoms

	H(1)							He(2)
1s	1.00							1.69
	Li(3)	Be(4)	B(5)	C(6)	N(7)	O(8)	F(9)	Ne(10)
1s	2.69	3.68	4.68	5.67	6.66	7.66	8.65	9.64
2s	1.28	1.91	2.58	3.22	3.85	4.49	5.13	5.76
2p			2.42	3.14	3.83	4.45	5.10	5.76

$$\epsilon_n \approx -\frac{[Z_{\text{eff}}(n)]^2}{n^2} \quad (\text{rydbergs})$$



Aufbau principle (构造原理) and electron configuration

Li atom

$$\psi = \psi(x_1, y_1, z_1, x_2, y_2, z_2, x_3, y_3, z_3)$$

The orbital approximation for atoms

$$\psi_{\text{atom}} = \varphi_{\alpha}(r_1)\varphi_{\beta}(r_2)\varphi_{\gamma}(r_3)$$

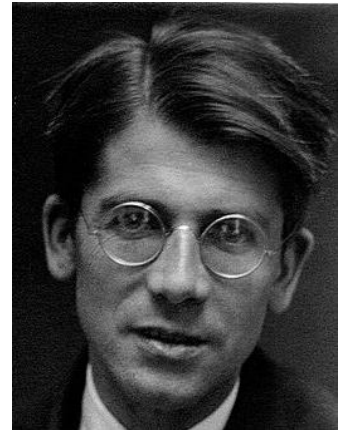
The **ground-state electronic configuration** of an atom (atomic number Z):
arrange the Hartree atomic orbitals in order of increasing energy, until all
Z electrons are in place.

The **Pauli exclusion principle**: no two electrons in an atom can have the same
set of four quantum numbers (n, ℓ , m, m_s).

Hund's rules: when electrons are added to Hartree orbitals of equal energy, **a
single electron enters each orbital** before a second one enters any orbital. In
addition, the lowest energy configuration is the one **with parallel spins**.

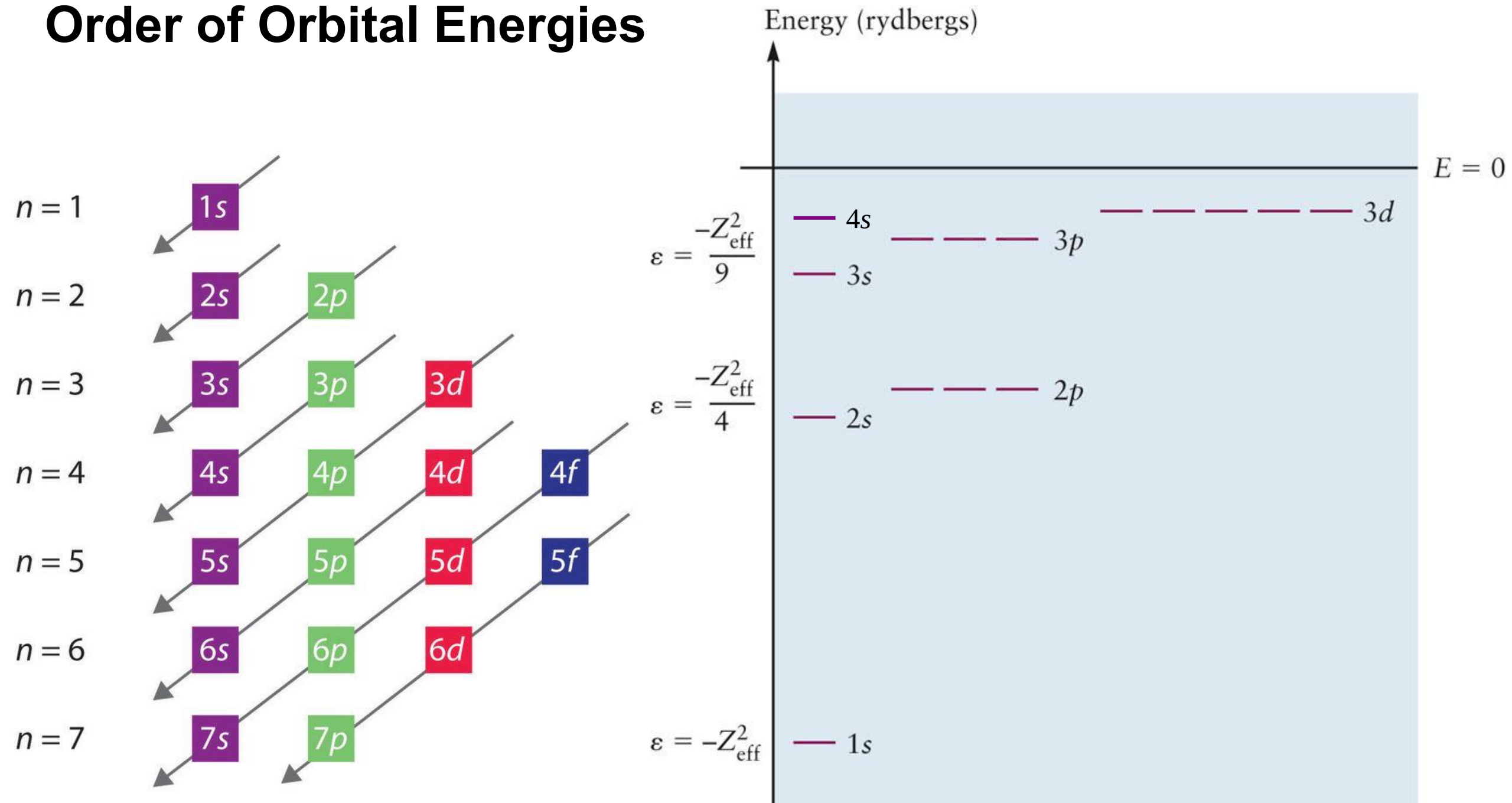


Wolfgang Pauli
(1900–1958)



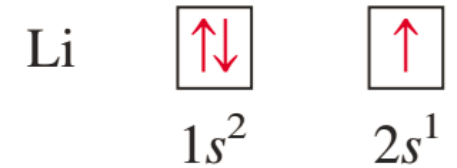
Friedrich Hund
(1896–1997)

Order of Orbital Energies



Electron Configurations of H thru Ar

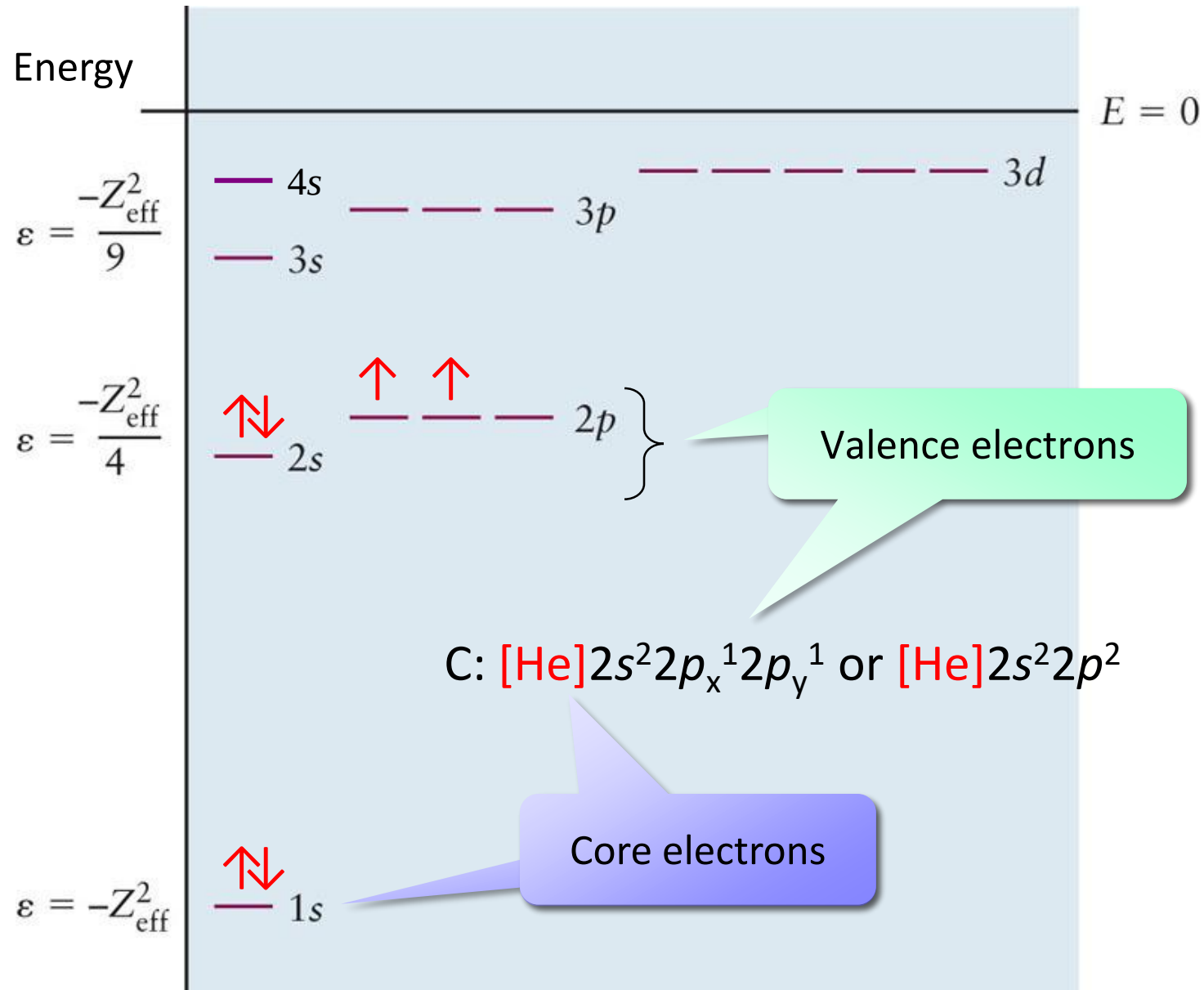
A substance is **paramagnetic (顺磁性)** if it is **attracted into a magnetic field**. Any substance that has **one or more unpaired electrons** in the atoms, molecules, or ions that compose it is paramagnetic because a net magnetic moment arises from each of the unpaired electrons.



A substance in which all the electrons are paired is weakly **diamagnetic (反磁性)**. It is **pushed out of a magnetic field**, although the force it experiences is much smaller in magnitude than the force that pulls a typical paramagnetic substance into a magnetic field.



Core and Valence Electrons



Transition-Metal Elements and Beyond

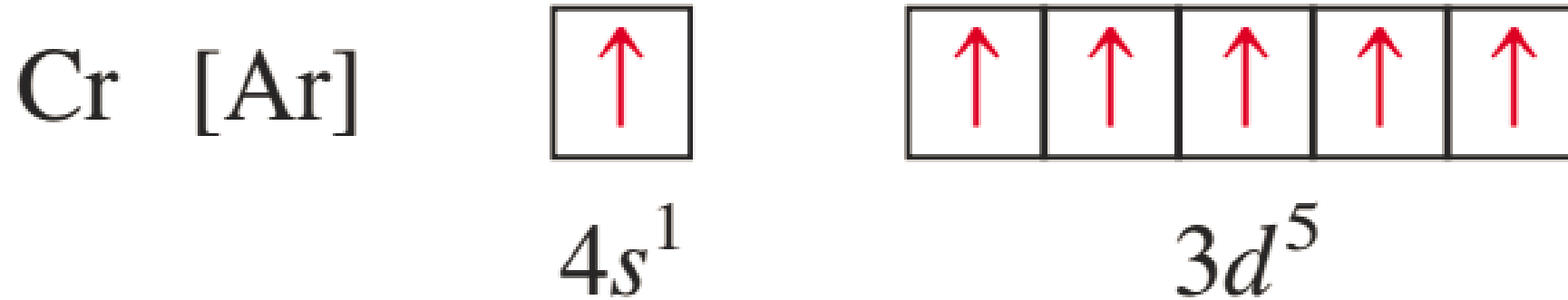
Advanced calculations for elements 19 (K) through 30 (Zn) predict that ϵ_{3d} and ϵ_{4s} are very close, so the build-up process becomes rather subtle.

Optical spectroscopy confirms that the ground state K is $[\text{Ar}]3d^04s^1$ and that of Ca is $[\text{Ar}]3d^04s^2$, as predicted by the sequence of calculated orbital energies $\epsilon_{3d} > \epsilon_{4s}$

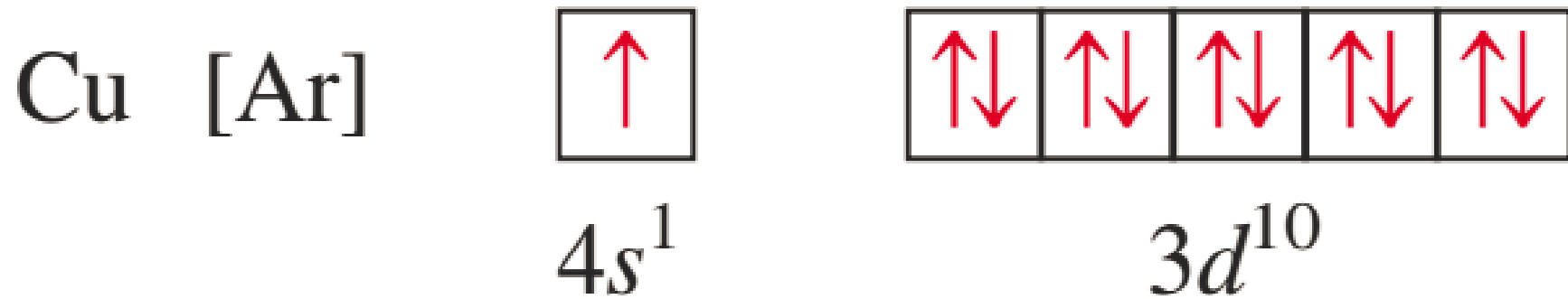
For Sc and the elements beyond, advanced calculations predict that $\epsilon_{3d} < \epsilon_{4s}$

Filling the 3d orbitals first would give the configurations $[\text{Ar}]3d^34s^0$ for Sc, $[\text{Ar}]3d^44s^0$ for Ti, and so on to $[\text{Ar}]3d^{10}4s^0$ for Ni according to calculation. However, experiments suggest $\epsilon_{3d} > \epsilon_{4s}$

Electron Configurations of K thru Zn



Two exceptions



Classification of groups of elements

1s																		1s	
H																		He	
2s–filling												2p–filling							
Li	Be											B	C	N	O	F	Ne		
3s–filling												3p–filling							
Na	Mg											Al	Si	P	S	Cl	Ar		
4s–filling		3d–filling										4p–filling							
K	Ca	Sc	Ti	V	Cr 3d⁵4s¹	Mn	Fe	Co	Ni	Cu 3d¹⁰4s¹	Zn	Ga	Ge	As	Se	Br	Kr		
5s–filling		4d–filling										5p–filling							
Rb	Sr	Y	Zr	Nb 4d⁴5s¹	Mo 4d⁵5s¹	Tc	Ru 4d⁷5s¹	Rh 4d⁸5s¹	Pd 4d¹⁰	Ag 4d¹⁰5s¹	Cd	In	Sn	Sb	Te	I	Xe		
6s–filling		5d–filling										6p–filling							
Cs	Ba	Lu	Hf	Ta	W	Re	Os	Ir	Pt 5d⁹6s¹	Au 5d¹⁰6s¹	Hg	Tl	Pb	Bi	Po	At	Rn		
7s–filling		6d–filling																	
Fr	Ra	Lr	Rf	Ha	Sg	Ns	Hs	Mt	Uun	Uuu									

4f–filling													
La 5d¹6s²	Ce 4f¹5d¹6s²	Pr	Nd	Pm	Sm	Eu	Gd 4f⁷5d¹6s²	Tb	Dy	Ho	Er	Tm	Yb

5f–filling													
Ac 6d¹7s²	Th 6d²7s²	Pa 5f²6d¹7s²	U 5f³6d¹7s²	Np 5f⁴6d¹7s²	Pu	Am	Cm 5f⁷6d¹7s²	Bk	Cf	Es	Fm	Md	No

Transition-Metal Elements and Beyond

In the sixth period, the filling of the 4f orbitals (and the generation of **the f-block elements**) begins as the rare-earth (lanthanide) elements from lanthanum to ytterbium are reached.

The configurations determined from calculations and experiment can be recalled as needed by assuming that the orbitals are filled in the sequence:

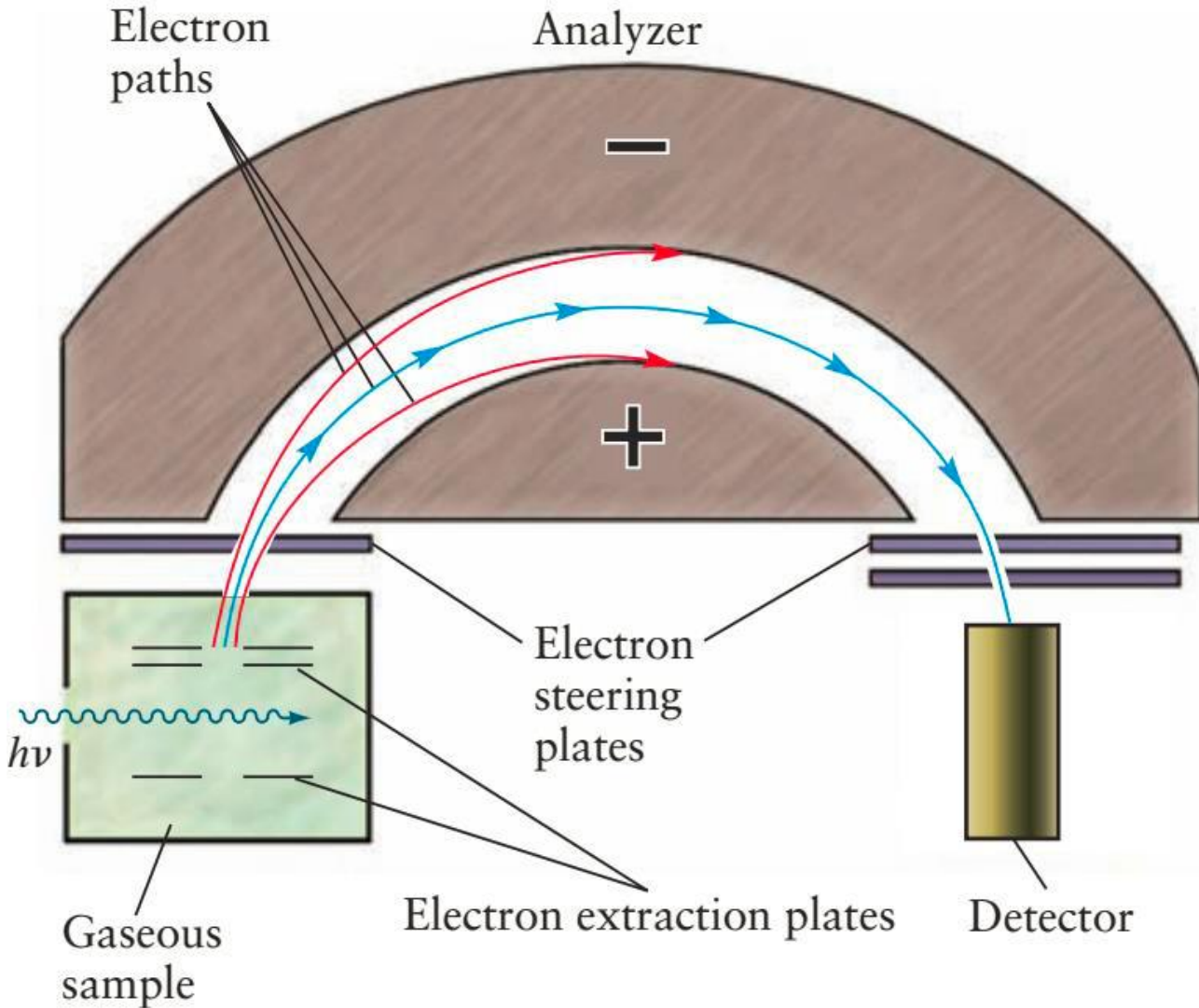
$1s \rightarrow 2s \rightarrow 2p \rightarrow 3s \rightarrow 3p \rightarrow 4s \rightarrow 3d \rightarrow 4p \rightarrow 5s \rightarrow 4d \rightarrow 5p \rightarrow 6s \rightarrow 4f \rightarrow 5d \rightarrow 6p \rightarrow 7s \rightarrow 5f \rightarrow 6d$.

The energies of the 4f, 5d, and 6s orbitals are **comparable over much of the sixth period**, and thus their order of filling is erratic.

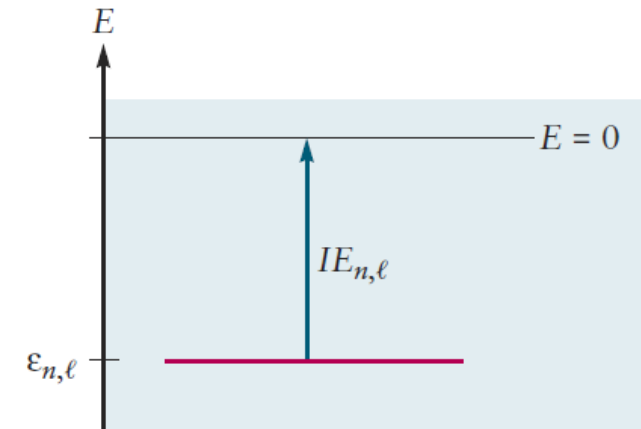
Major points of today's lecture

- **Quantum mechanics and atomic structure**
 1. The hydrogen atom
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Photoelectron Spectroscopy (PES)



$$IE = h\nu_{\text{photon}} - \frac{1}{2} m_e v_{\text{electron}}^2$$



The ionization energy $IE_{n,\ell}$ is the energy required to remove an electron from this atomic state and set it free from the atom.

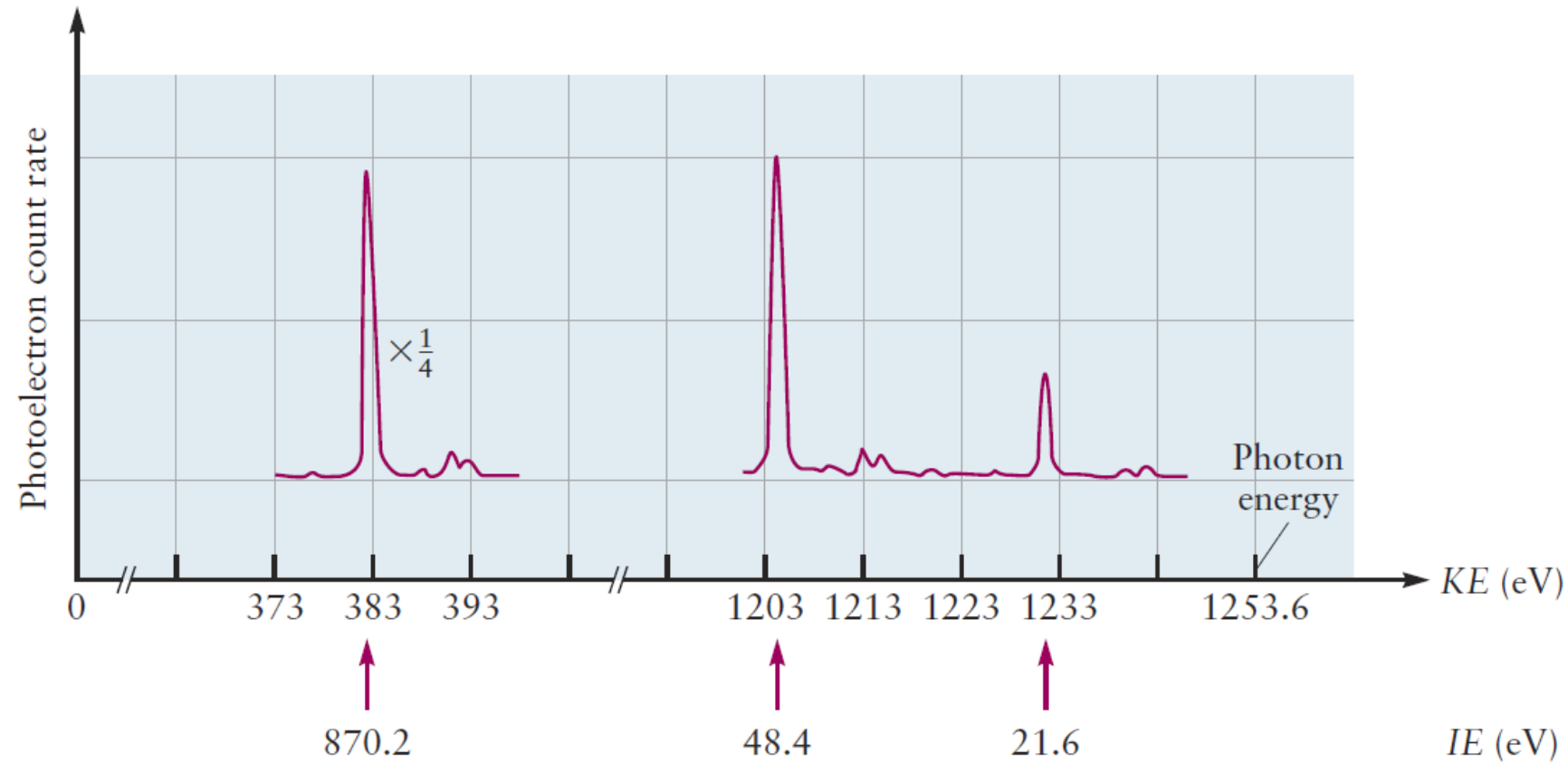
$$IE_{\alpha} = -\epsilon_{\alpha}$$

Koopmans's (科普曼斯) approximation: the ionization energy of an electron is the negative of the energy of the Hartree orbital of the electron.

It is **not strictly valid** because it assumes the orbital energies are the same in the ion as in the parent atom, despite the loss of an electron.

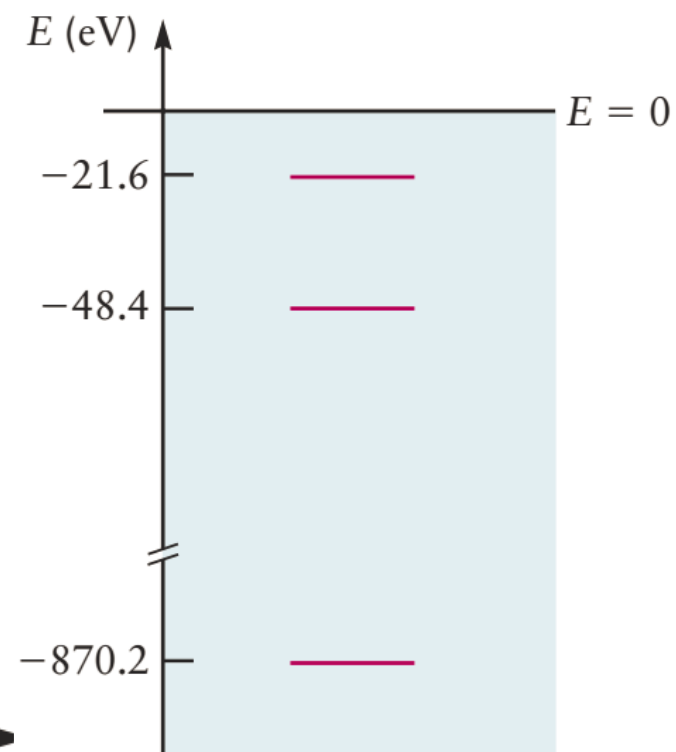
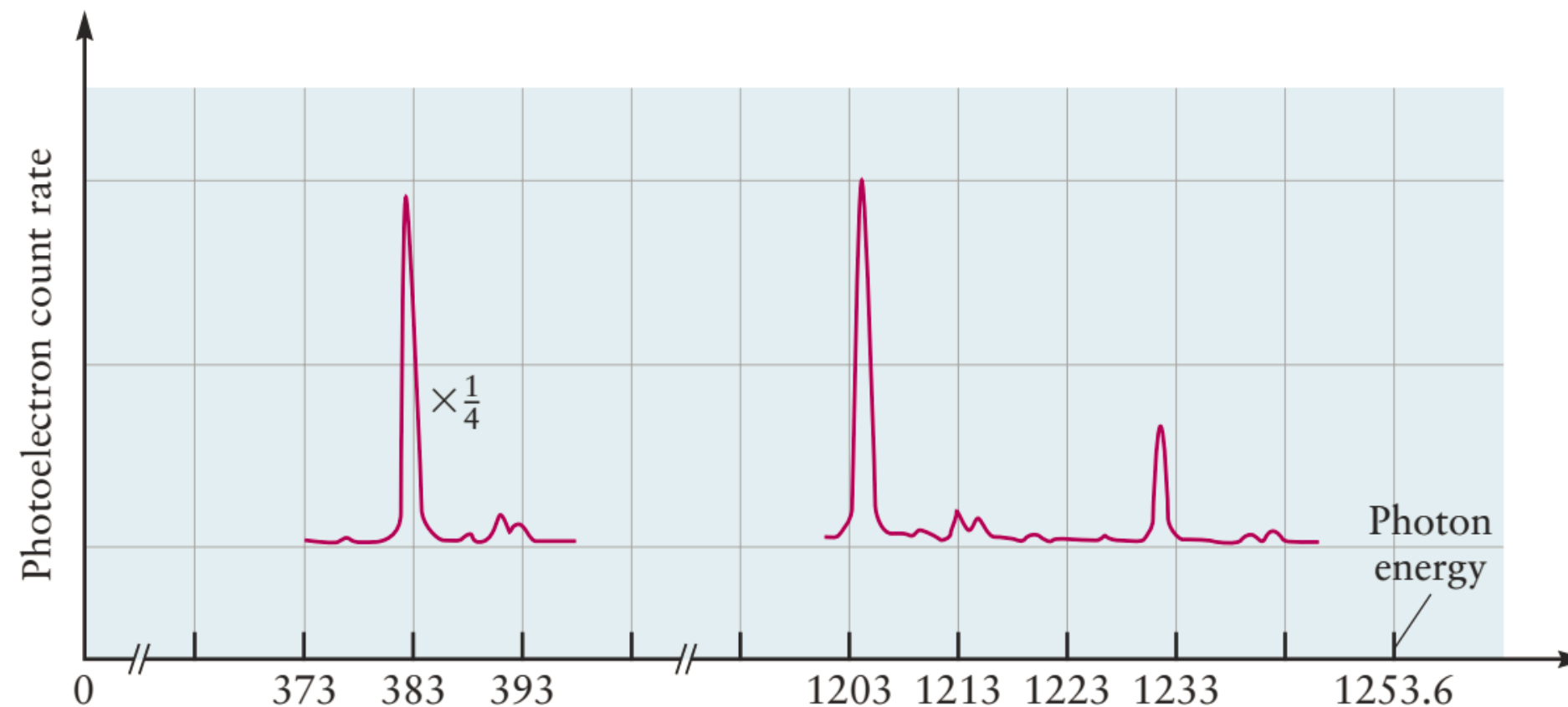
Construct the energy-level diagram for neon from the data in Figure 5.23.

FIGURE 5.23 Photoelectron spectrum of neon. The spectrum shows three peaks, demonstrating that the electrons of neon are organized in three bonding states of distinct energy values. The peak at 383.4 eV has been reduced by a factor of 4 for display on the same scale as the other two.

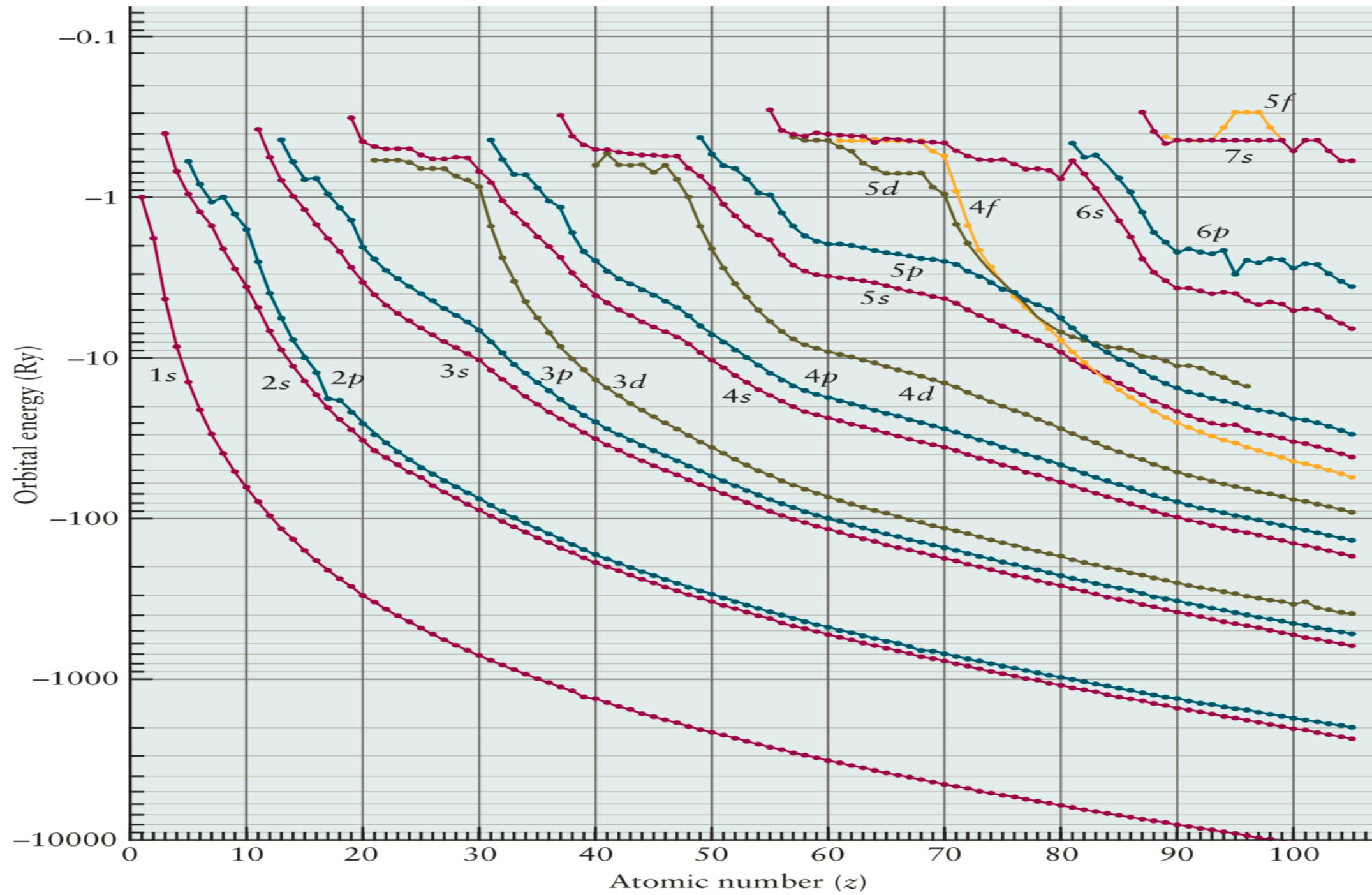


Photoelectron Spectroscopy (PES)

Ne: $1s^2, 2s^2, 2p^6$

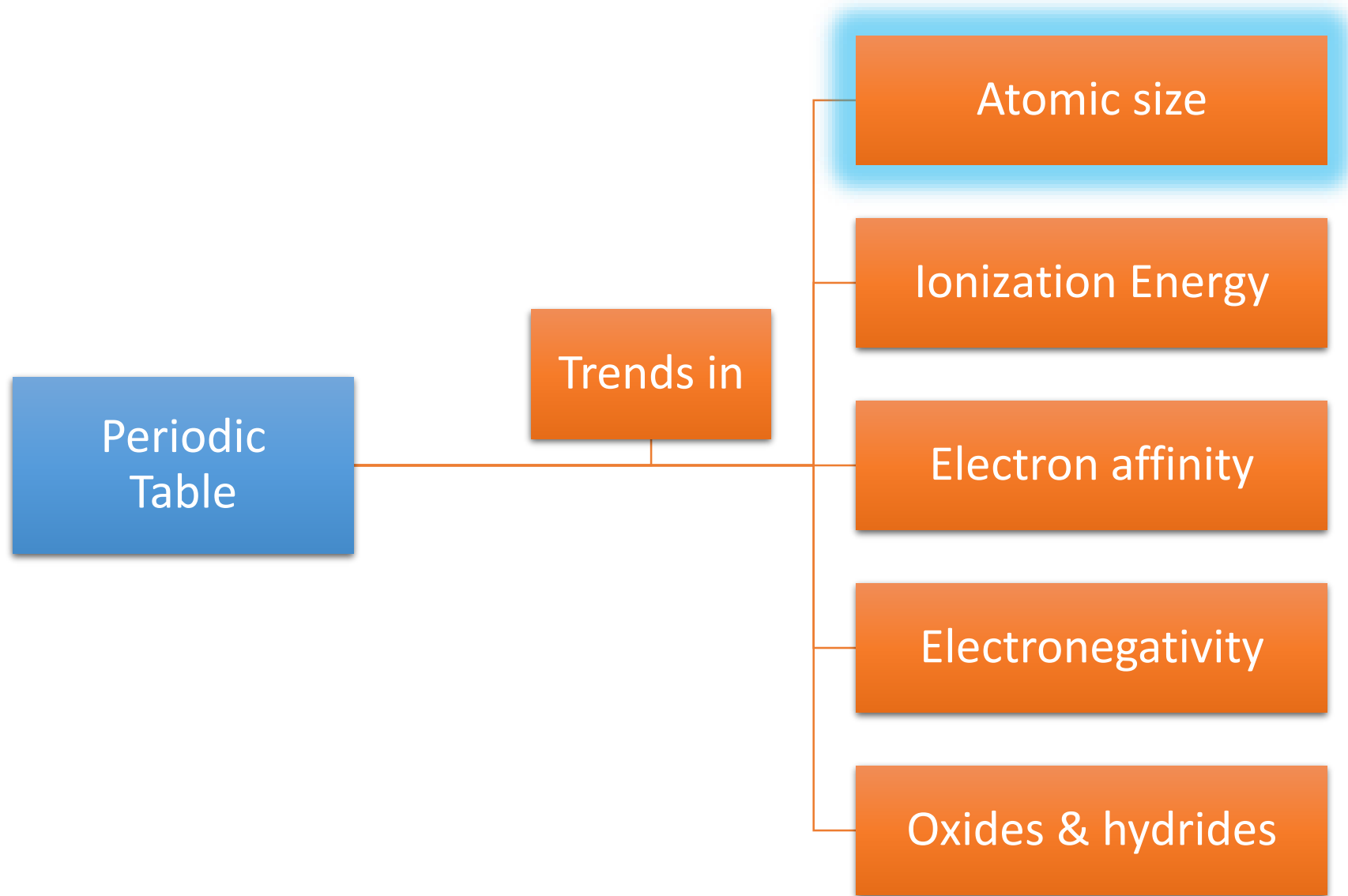


Kinetic Energy (eV)	Ionization Energy (eV)
383.4	870.2
1205.2	48.4
1232.0	21.6



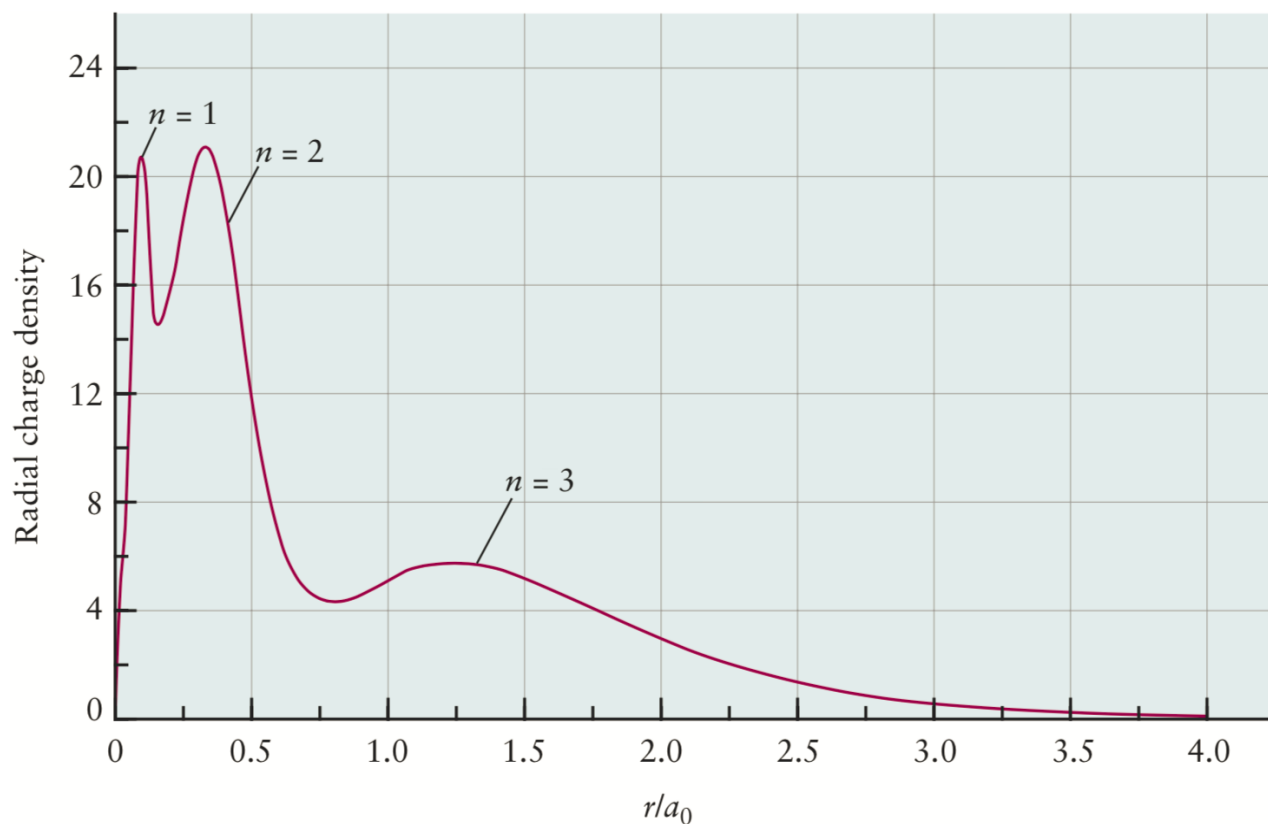
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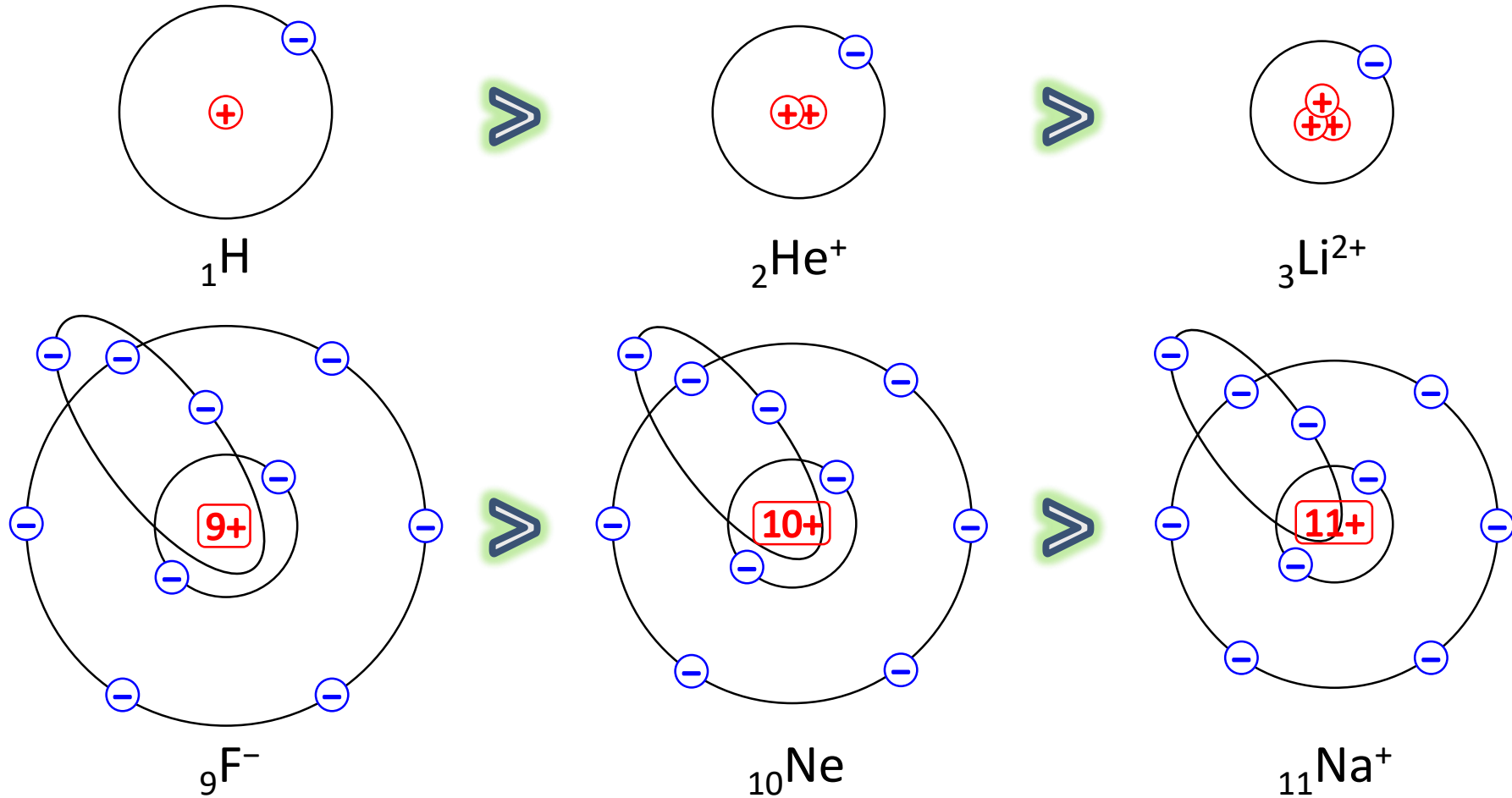
Sizes of atoms and ions

Atomic Radius (原子半径): not a precisely defined concept, these sizes can be estimated in several ways
根据不同的标度和测量方法，原子半径的定义不同：范德华半径，共价半径，金属半径等。同一原子依不同定义得到的原子半径差别可能很大。



原子半径： 主要受**电子层数**和**核电荷数**两个因素影响。一般来说，电子层数越多，核电荷数越小，原子半径越大

Radii of Isoelectronic (等电子的) Ions



Atomic/ionic radius = radius of **outermost orbital**

Radii of Atoms with Same n

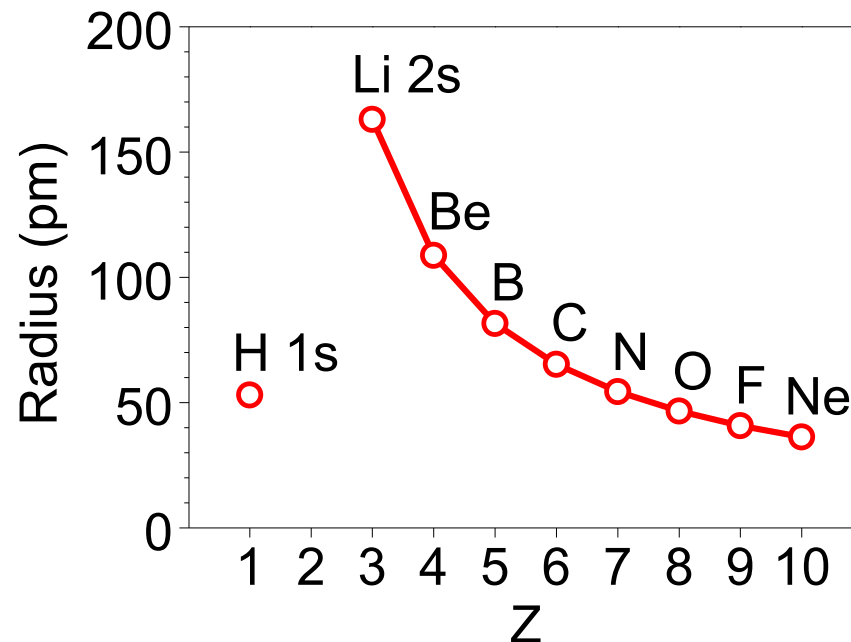
Atom	Z	$Z_{\text{eff}}(2s)$	Valence shell
Li	3	~ 1.30	$2s^1$
Be	4	~ 1.95	$2s^2$
B	5	~ 2.60	$2s^2 2p^1$
C	6	~ 3.25	$2s^2 2p^2$

For s and p subshells,

$$\Delta Z_{\text{eff}}(ns) \approx 0.65 \Delta Z$$

For 2s and 2p,

$$Z_{\text{eff}} \approx 0.65(Z - 1)$$



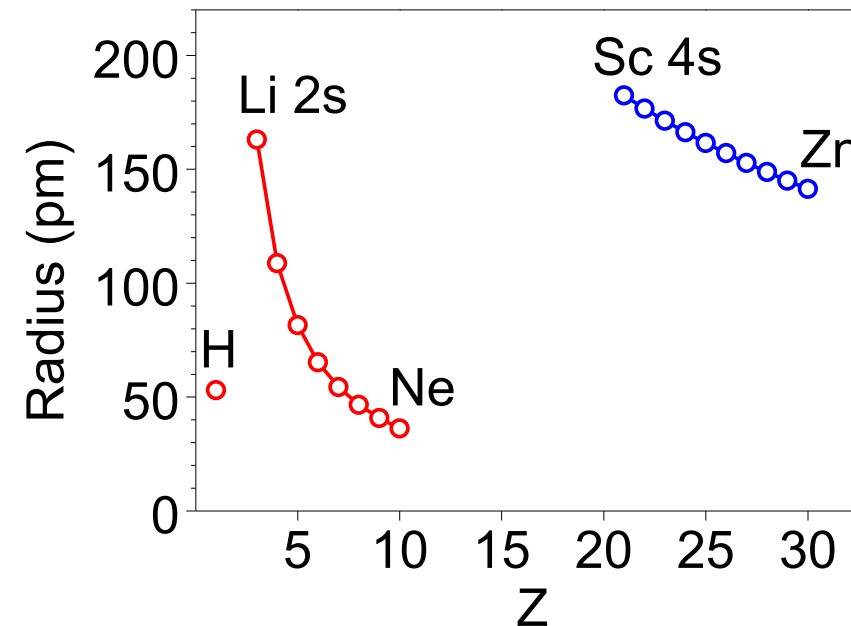
John C. Slater
(MIT, 1900–1976)

Radii of Atoms with Same n

Atom	Z	$Z_{\text{eff}}(4s)$	Valence shell
Mn	25	5.28	$3d^5 4s^2$
Fe	26	5.43	$3d^6 4s^2$
Co	27	5.58	$3d^7 4s^2$
Ni	28	5.71	$3d^8 4s^2$

Slater's Rules
(1930)
OGC7 p.214

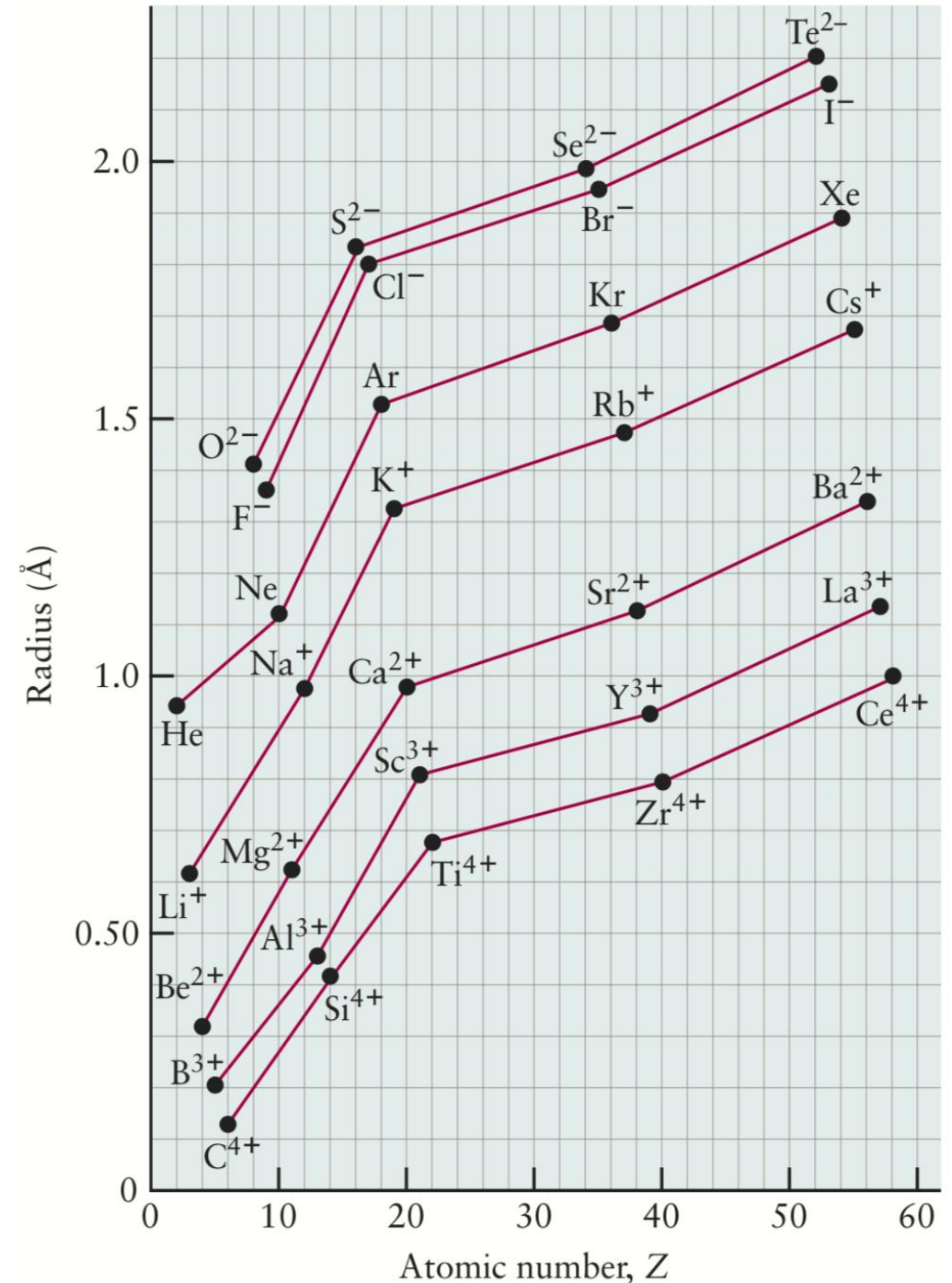
For d subshells, $\Delta Z_{\text{eff}}(ns) \approx 0.15 \Delta Z$;



Sizes of atoms and ions

Elements/ions in the same group of the periodic table: the radius usually increases with increasing atomic number.

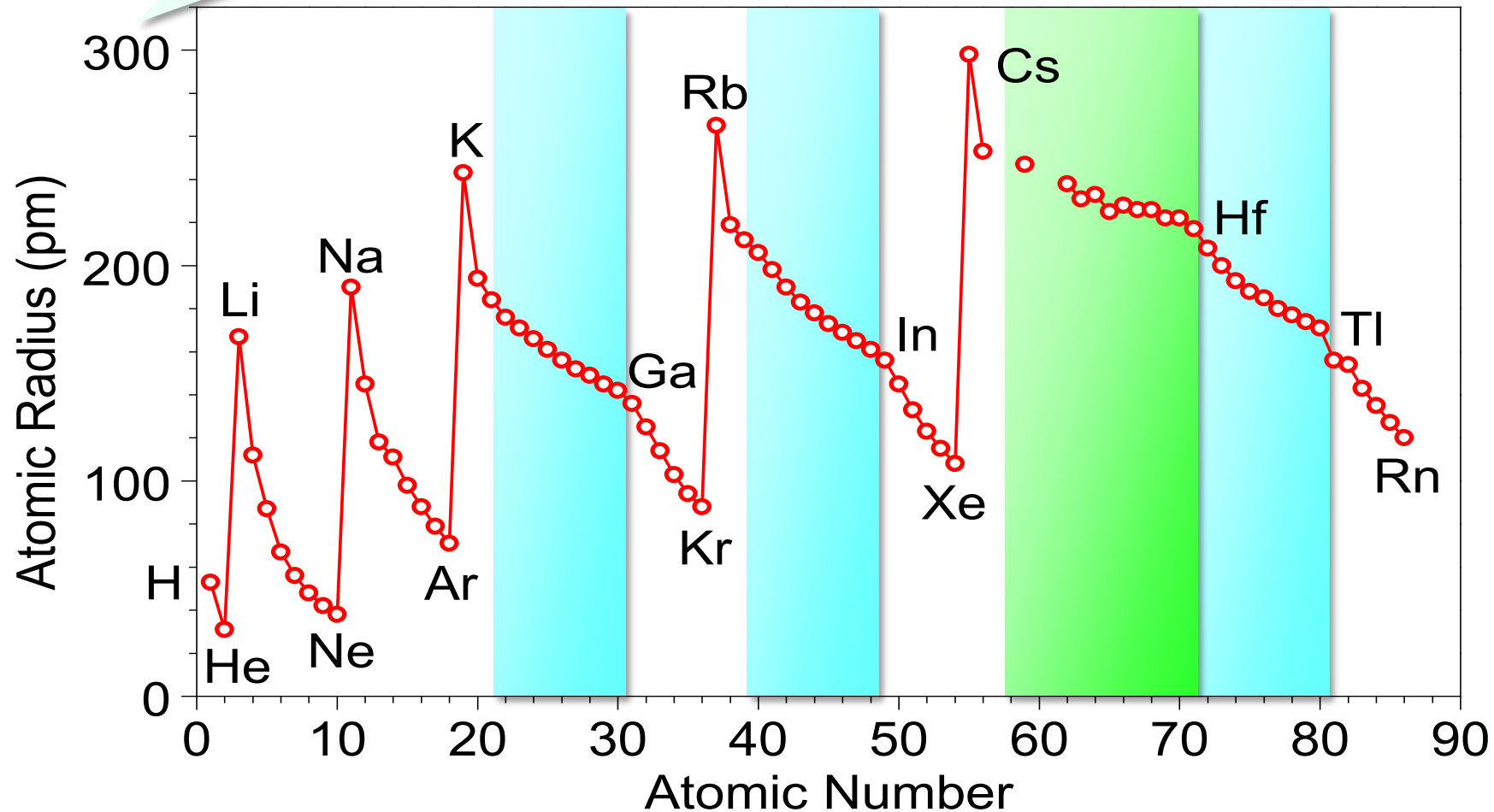
Elements/ions in the same period of the periodic table: the radius usually decreases with increasing atomic number.



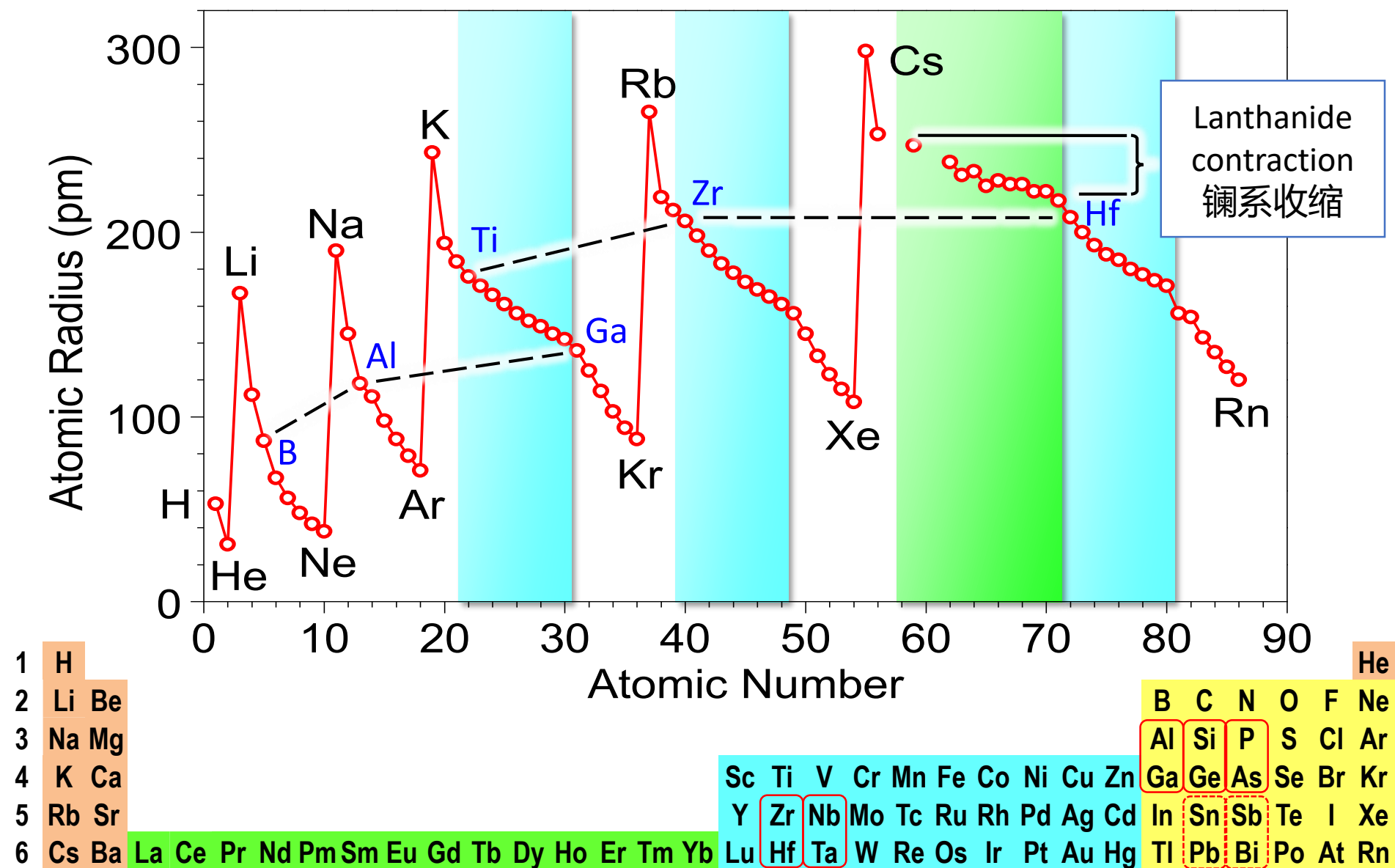
Trends in Atomic Radii (1)

$$\bar{r}_{n\ell} \approx \frac{n^2 a_0}{Z_{\text{eff}}(n)} \left\{ 1 + \frac{1}{2} \left[1 - \frac{\ell(\ell + 1)}{n^2} \right] \right\}$$

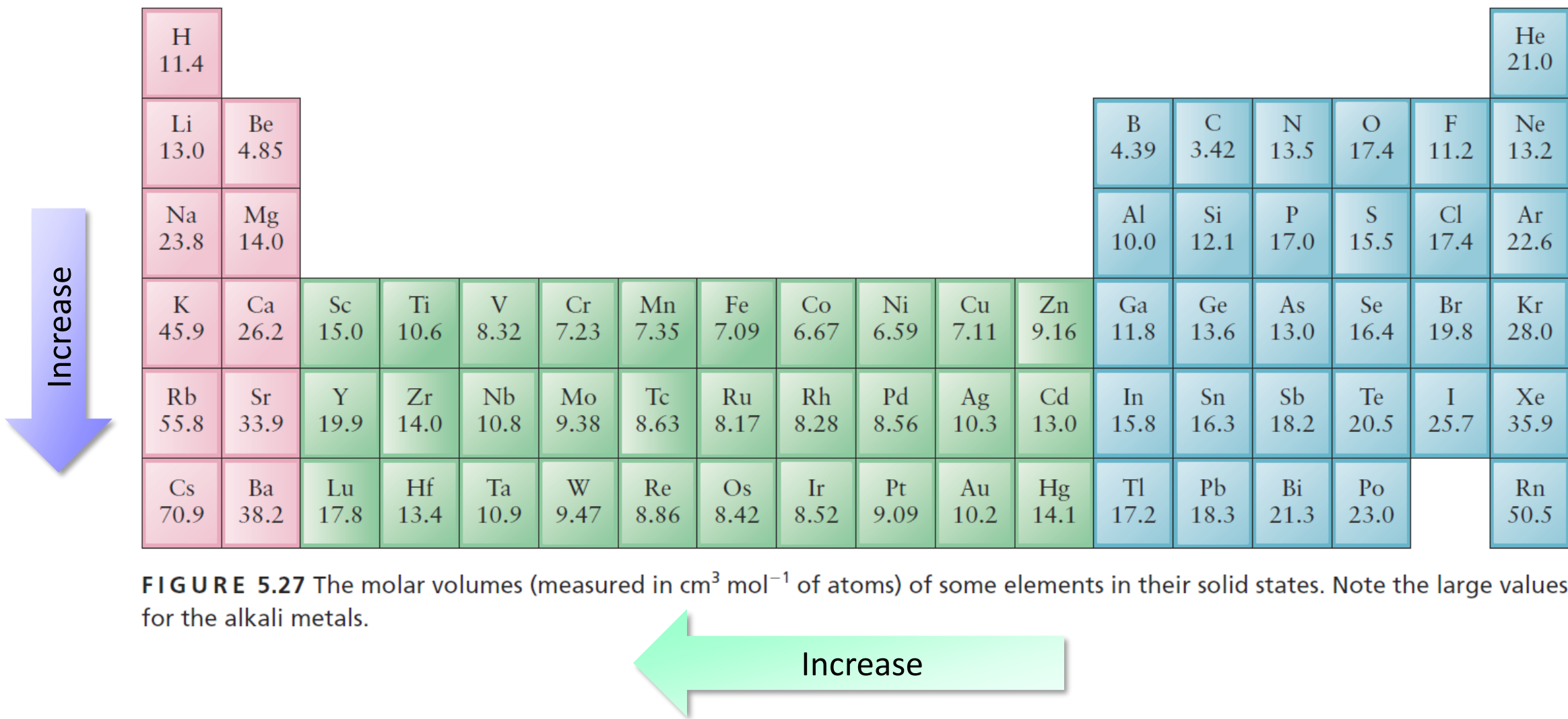
Calculated values



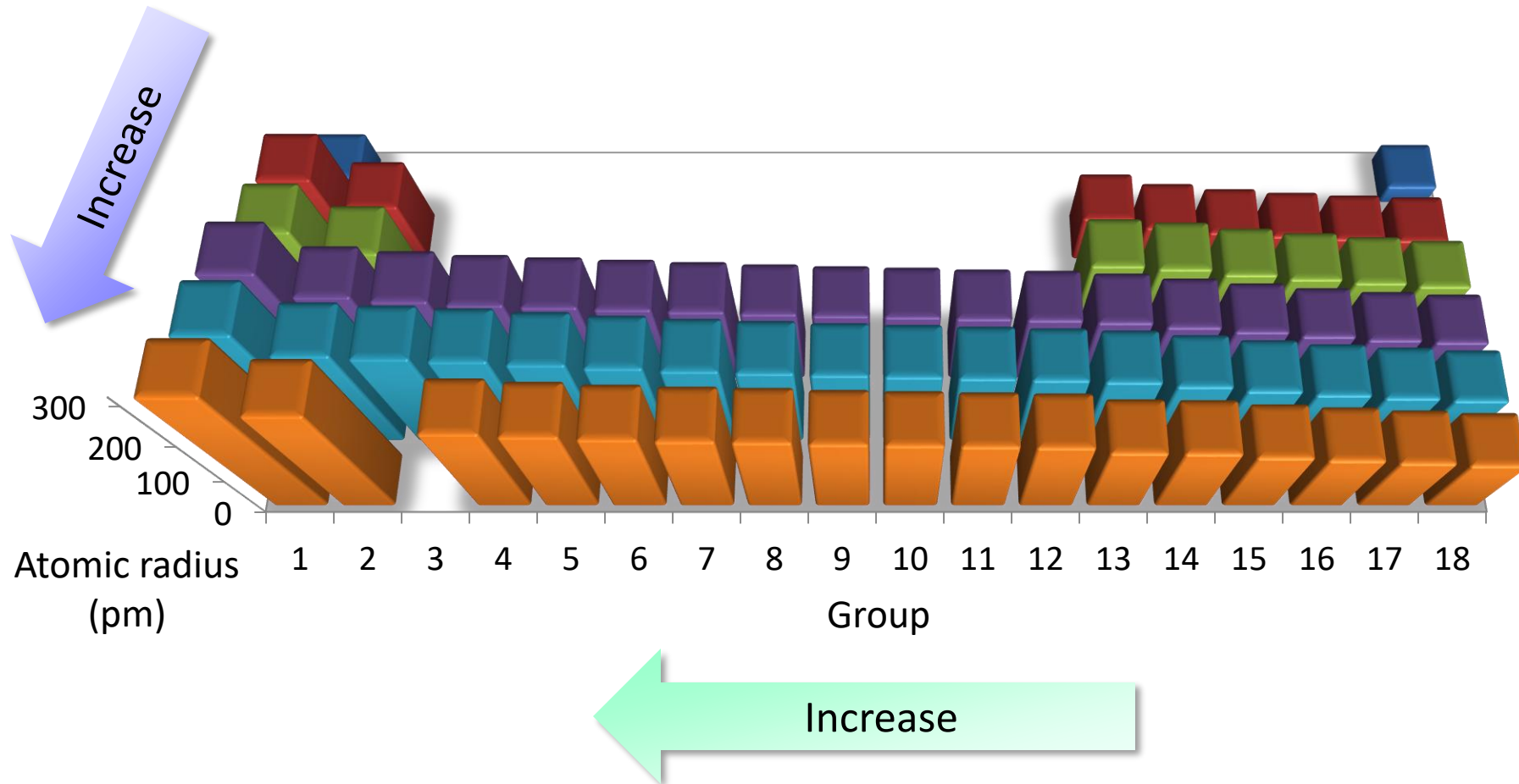
Trends in Atomic Radii (2)



Sizes of atoms and ions



Periodic Trends in Atomic Radii



The Periodic Table

		Group 族 →																IUPAC #		
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	
		IA	IIA	IIIB	IVB	VB	VIB	VII B	VIII B		IB		IIB	IIIA	IVA	VA	VIA	VIIA	VIIIA	
Period 周期 ↓	1	H			s block		p block													He
	2	Li	Be		d block		f block								B	C	N	O	F	Ne
	3	Na	Mg												Al	Si	P	S	Cl	Ar
	4	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr	
	5	Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe	
	6	Cs	Ba	*	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn	
	7	Fr	Ra	**																
		*	La Ce Pr Nd Pm Sm Eu Gd Tb Dy Ho Er Tm Yb Lu																	
		**	Ac Th Pa U Np Pu Am Cm Bk Cf Es Fm Md No Lr																	

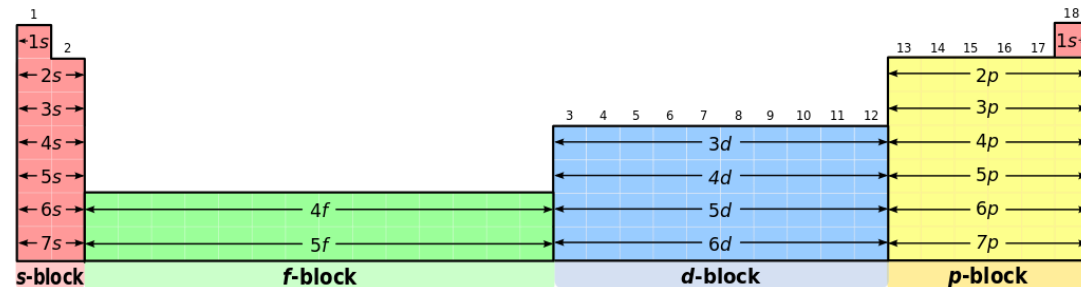
Periodic Trends in Atomic Radii: summary

- Atomic radius is determined by the valence s subshell
- When estimating Z_{eff} for ns :
 - np electrons only partly screens ns ;
 - $(n-1)d$ electrons efficiently screens ns ;
 - $(n-2)f$ electrons completely screens ns .

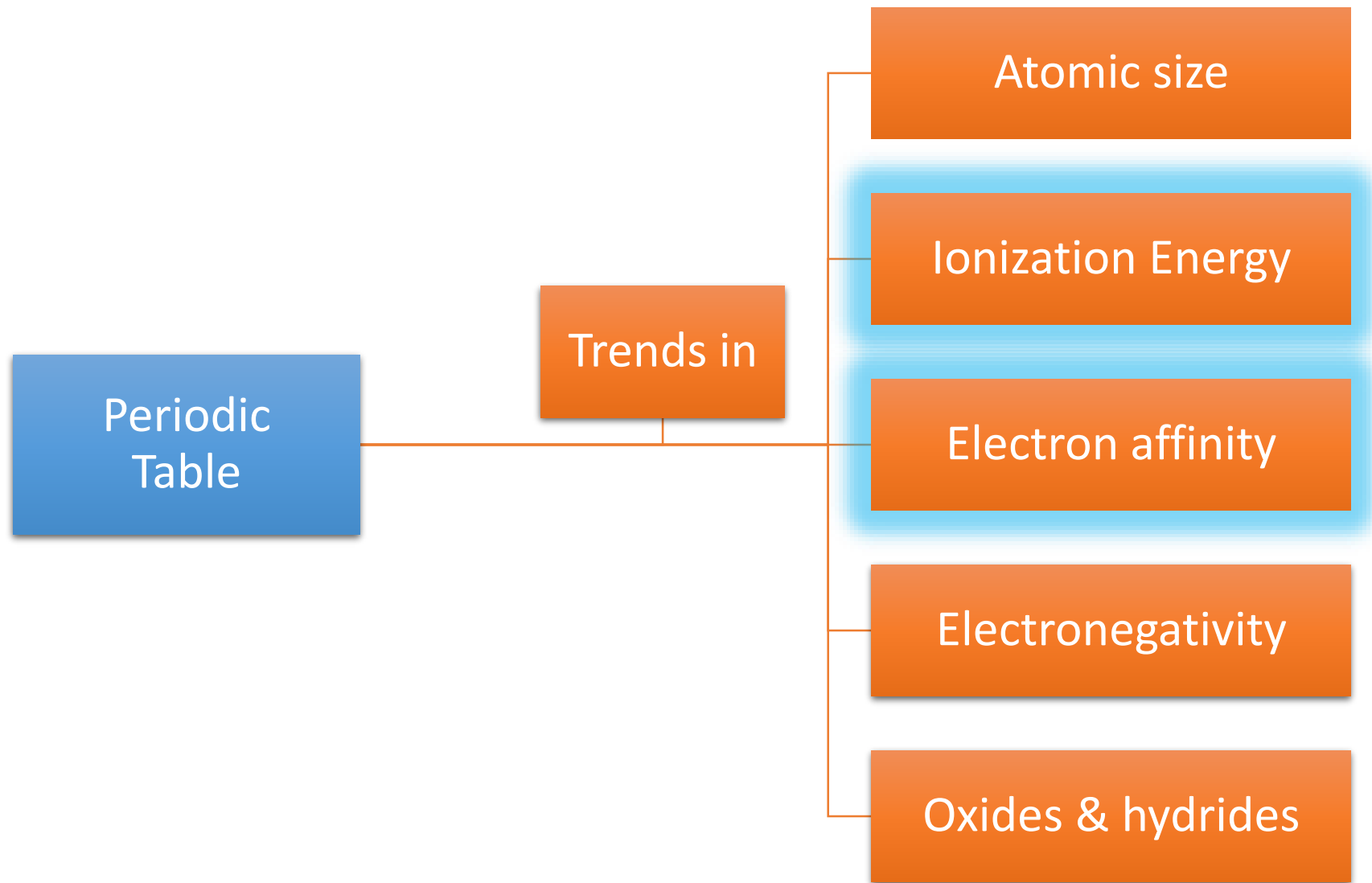
- The trends in atomic radius $r_{n,Z}$
 - Periods according to n
 - $r_{n,Z}$ decreases within the same period
 - s -, p -, and d -blocks

$$r_{n,Z} \approx \frac{n^2}{Z_{\text{eff}}} a_0$$

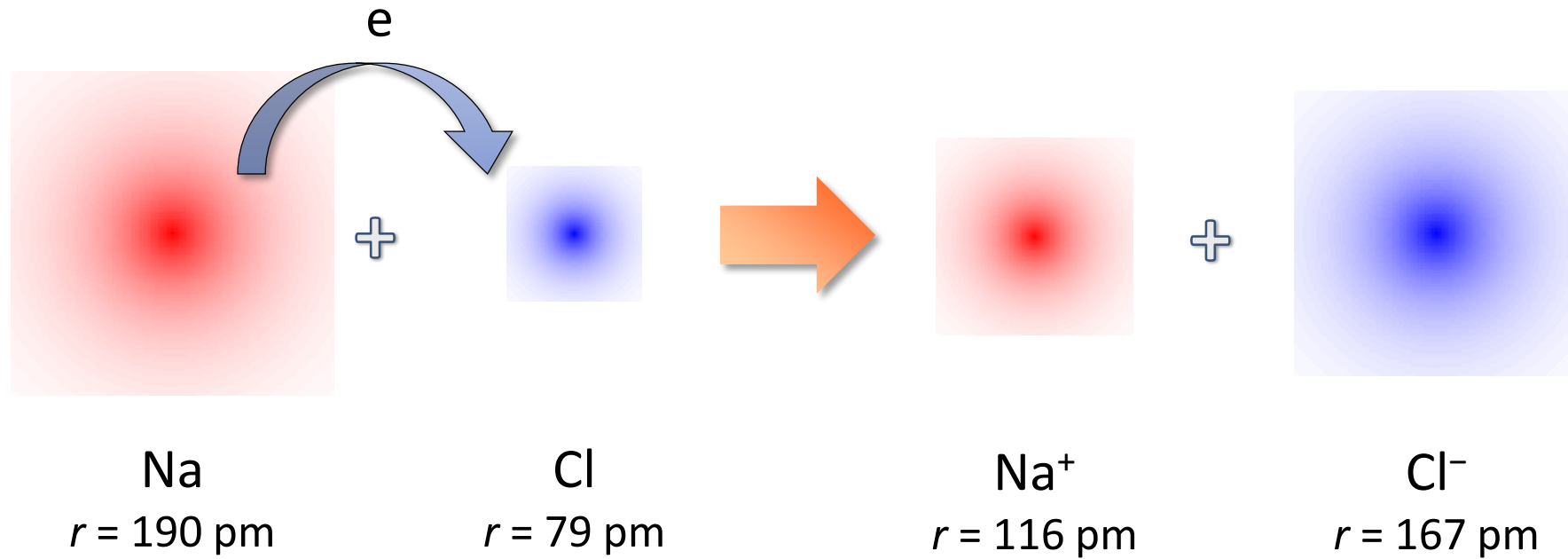
- The Periodic Table



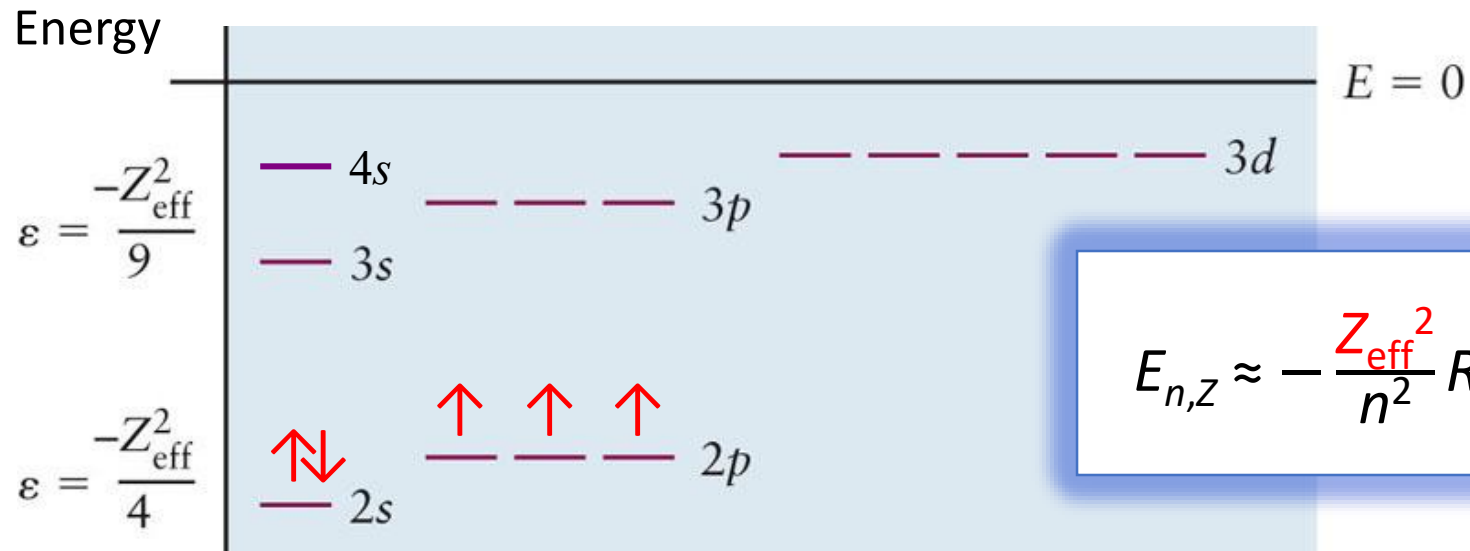
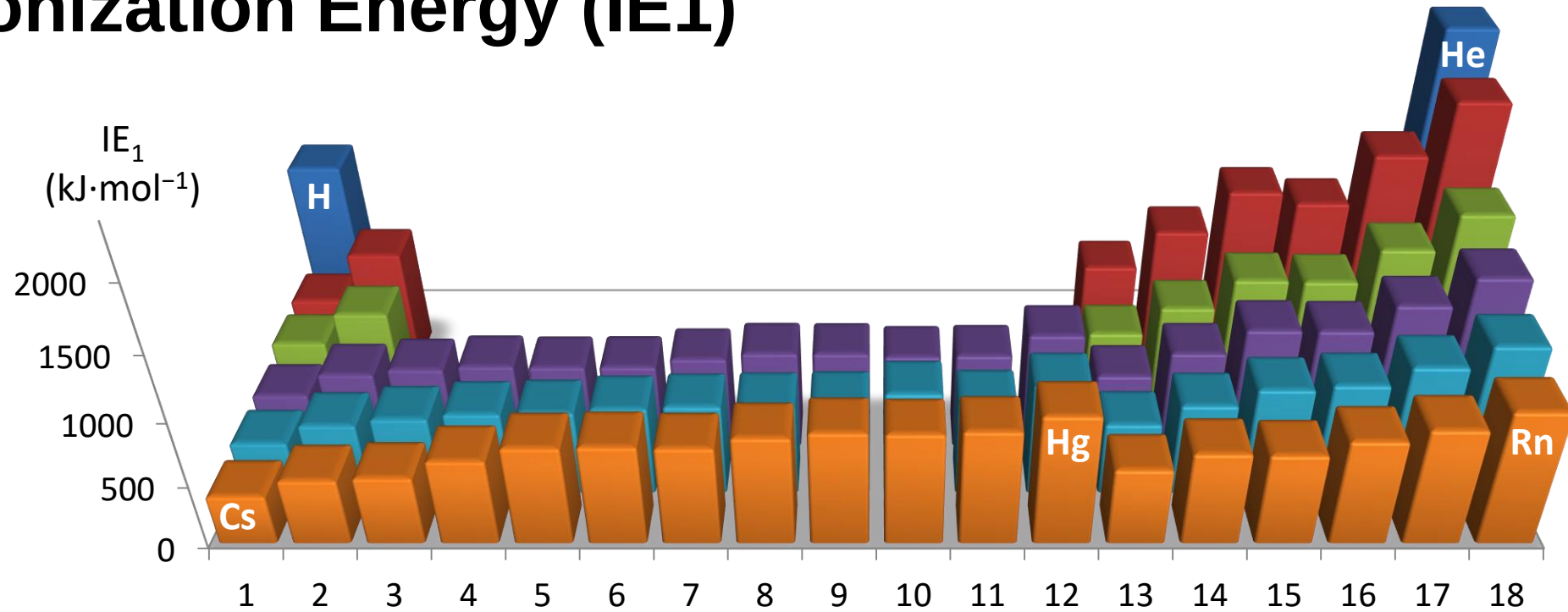
Predict which atom or ion in each of the following pairs should be larger: (a) Kr or Rb, (b) Y or Cd, (c) F^- or Br^- .



Gaining and Losing an Electron

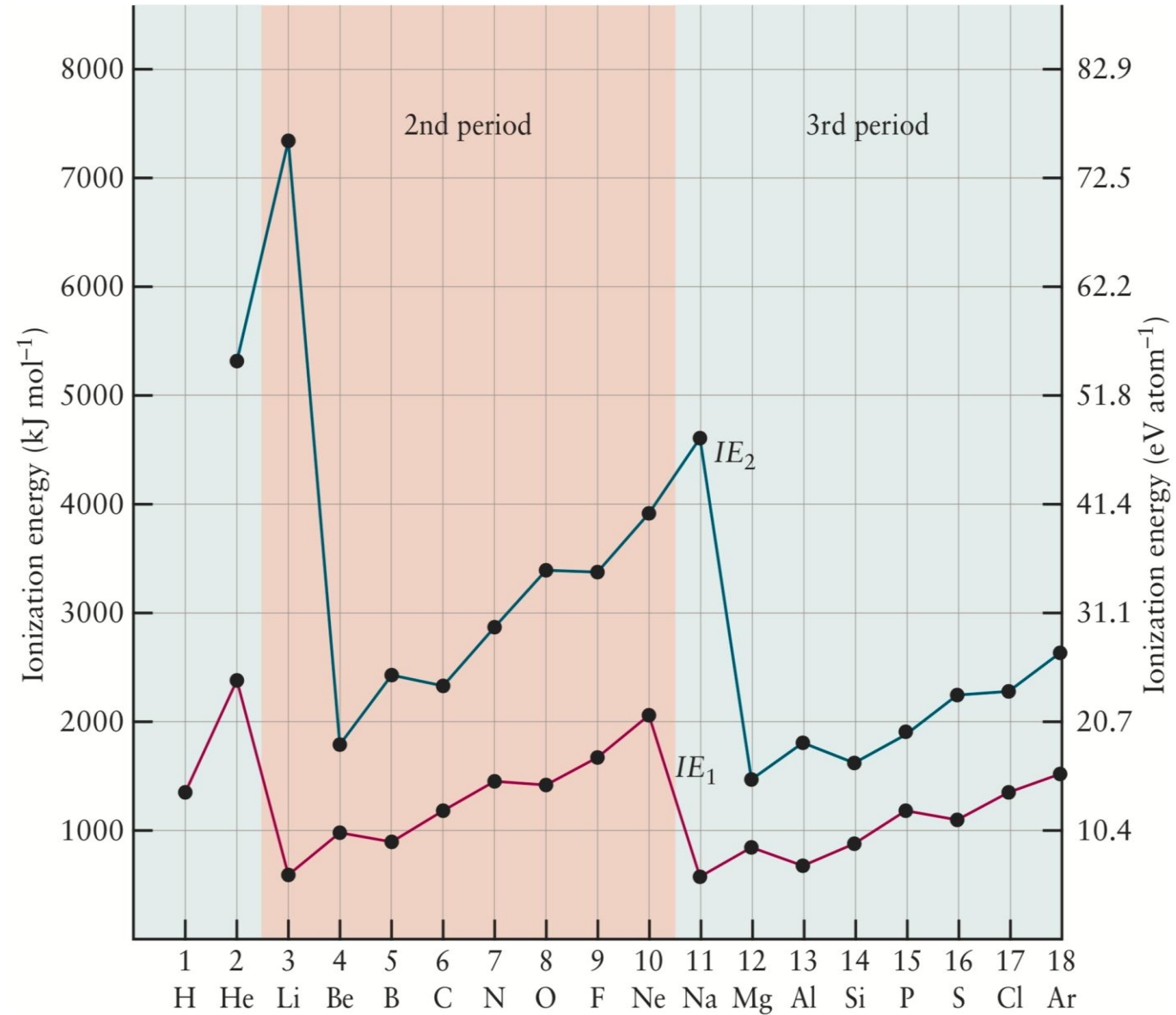


First Ionization Energy (IE1)

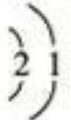
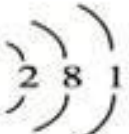


$$E_{n,Z} \approx -\frac{Z_{\text{eff}}^2}{n^2} R_H$$

Periodic Trends in Ionization Energies



Sizes of atoms and ions

元素名称	元素符号	核电荷数	电子层结构	电子层数	原子半径	原子失去电子	金属性	与非金属、水反应
锂	Li	3		逐渐增多 ↓	逐渐增大 ↓	越来越容易 ↓	越来越强 ↓	越来越剧烈 ↓
钠	Na	11						
钾	K	19						
铷	Rb	37						
铯	Cs	55						

Anomaly of the Trend in IE

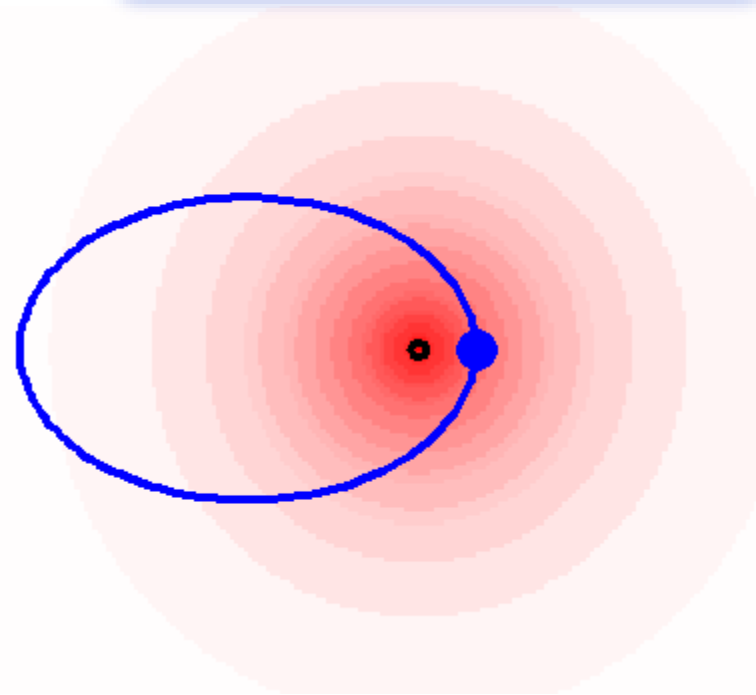
Relativistic effect of 6s electrons

$$v \rightarrow c \Rightarrow m_e' = \frac{m_e}{\sqrt{1-v^2/c^2}} \uparrow$$
$$\Rightarrow \text{Ry} = \frac{e^4 m_e}{8\epsilon_0^2 h^2} \uparrow \Rightarrow E_{n,Z} \approx -\frac{Z_{\text{eff}}^2}{n^2} \text{Ry} \downarrow$$

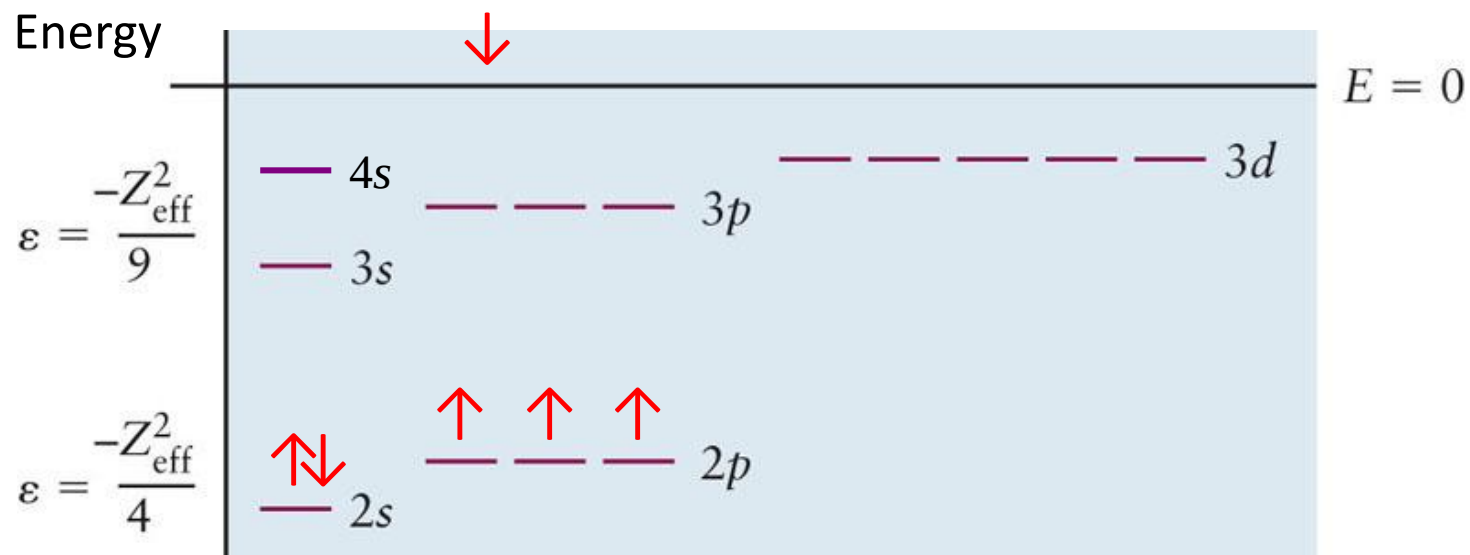
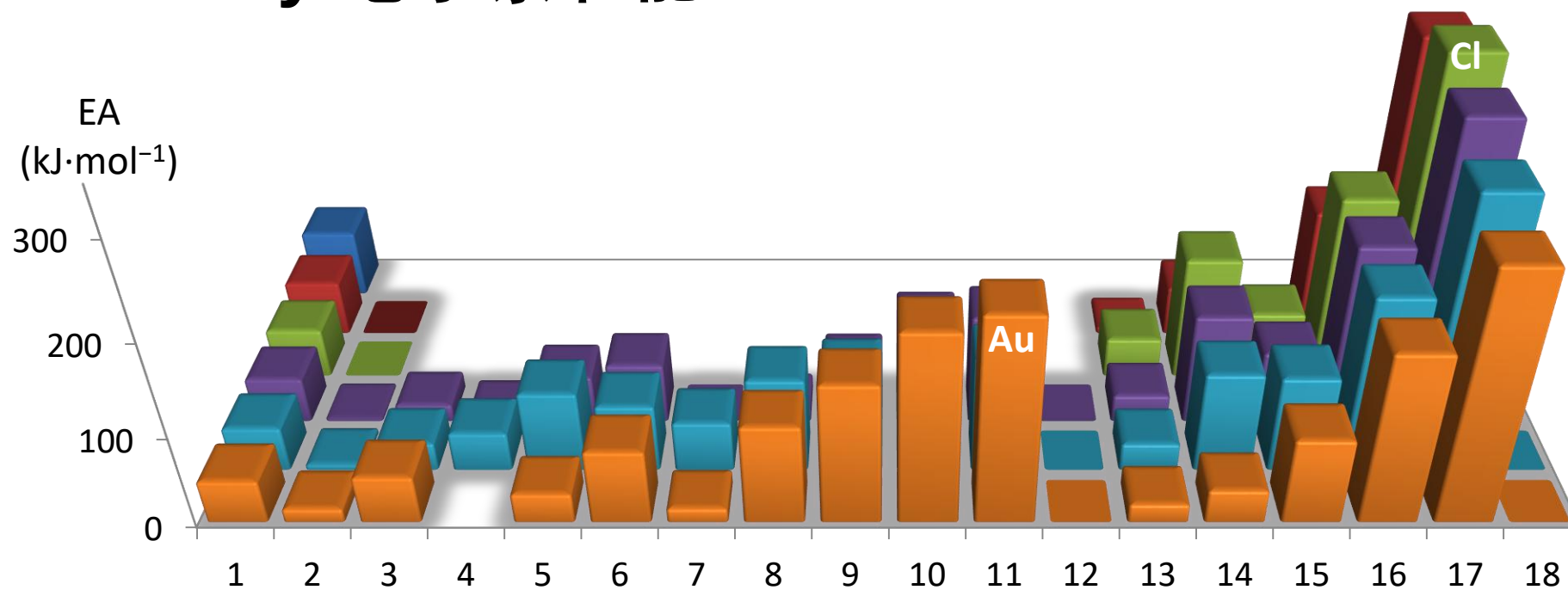
Electron configurations

- Hg: [Xe] $4f^{14} 5d^{10} 6s^2$
- Hg⁺: [Xe] $4f^{14} 5d^{10} 6s^1$
- Tl: [Xe] $4f^{14} 5d^{10} 6s^2 6p^1$

For $_{80}\text{Hg } 6s^2$,
 $v/c \approx 80/137 \approx 0.58$
 $m_e' = 1.23m_e$



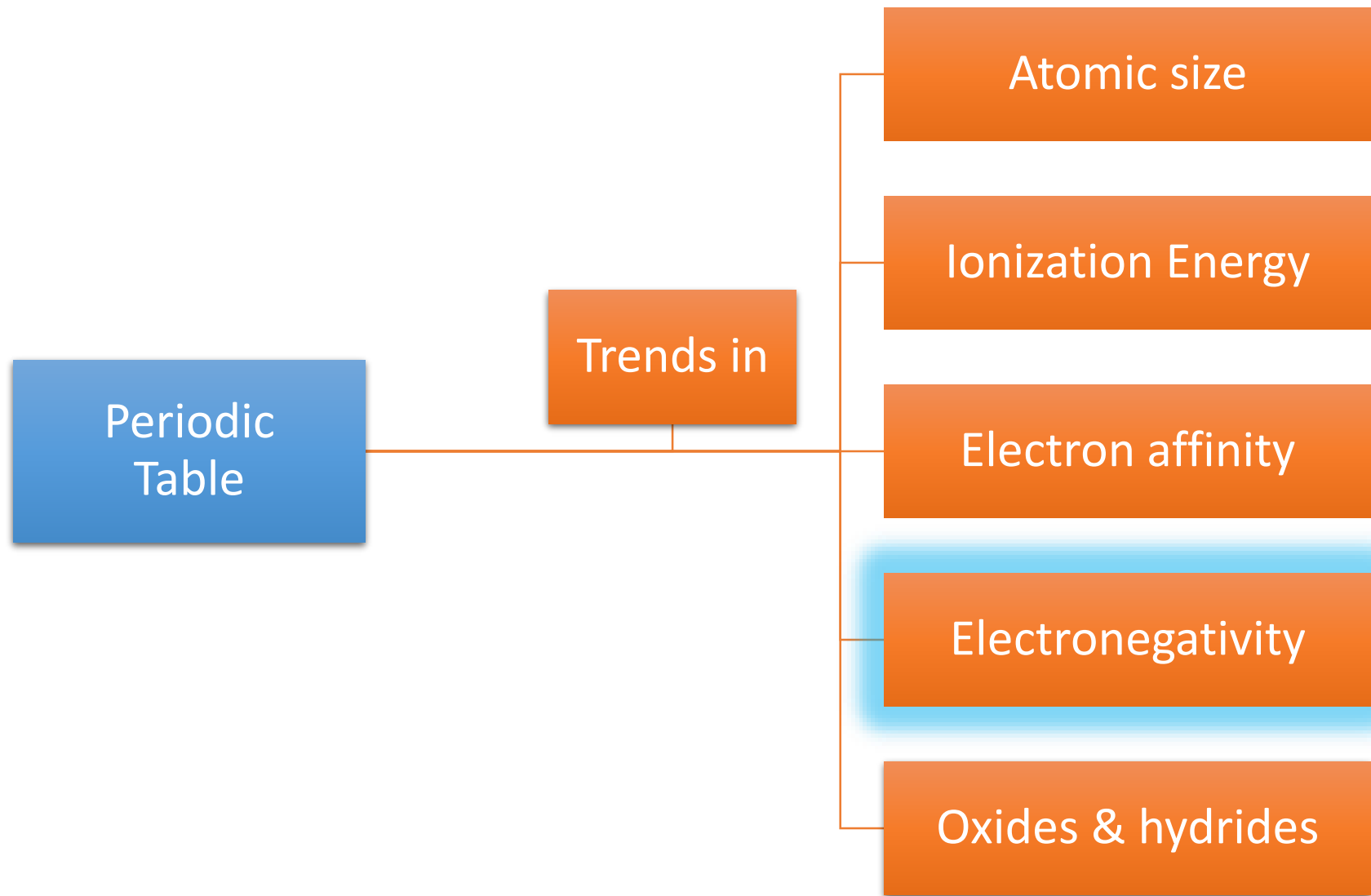
Electron Affinity 电子亲和能



Periodic Trends in Electron Affinity

H 73																He *	
Li 60	Be *											B 27	C 122	N *	O 141	F 328	Ne *
Na 53	Mg *											Al 43	Si 134	P 72	S 200	Cl 349	Ar *
K 48	Ca 2	Sc 18	Ti 8	V 51	Cr 64	Mn *	Fe 16	Co 64	Ni 111	Cu 118	Zn *	Ga 29	Ge 116	As 78	Se 195	Br 325	Kr *
Rb 47	Sr 5	Y 30	Zr 41	Nb 86	Mo 72	Tc 53	Ru 99	Rh 110	Pd 52	Ag 126	Cd *	In 29	Sn 116	Sb 103	Te 190	I 295	Xe *
Cs 46	Ba 14	Lu 50	Hf *	Ta 31	W 79	Re 14	Os 106	Ir 151	Pt 214	Au 223	Hg *	Tl 19	Pb 35	Bi 91	Po 183	At 270	Rn *

FIGURE 5.29 Electron affinities (measured in kJ mol^{-1}) of gaseous atoms of the elements. An asterisk means that the element does not have a stable anion in the gas phase.

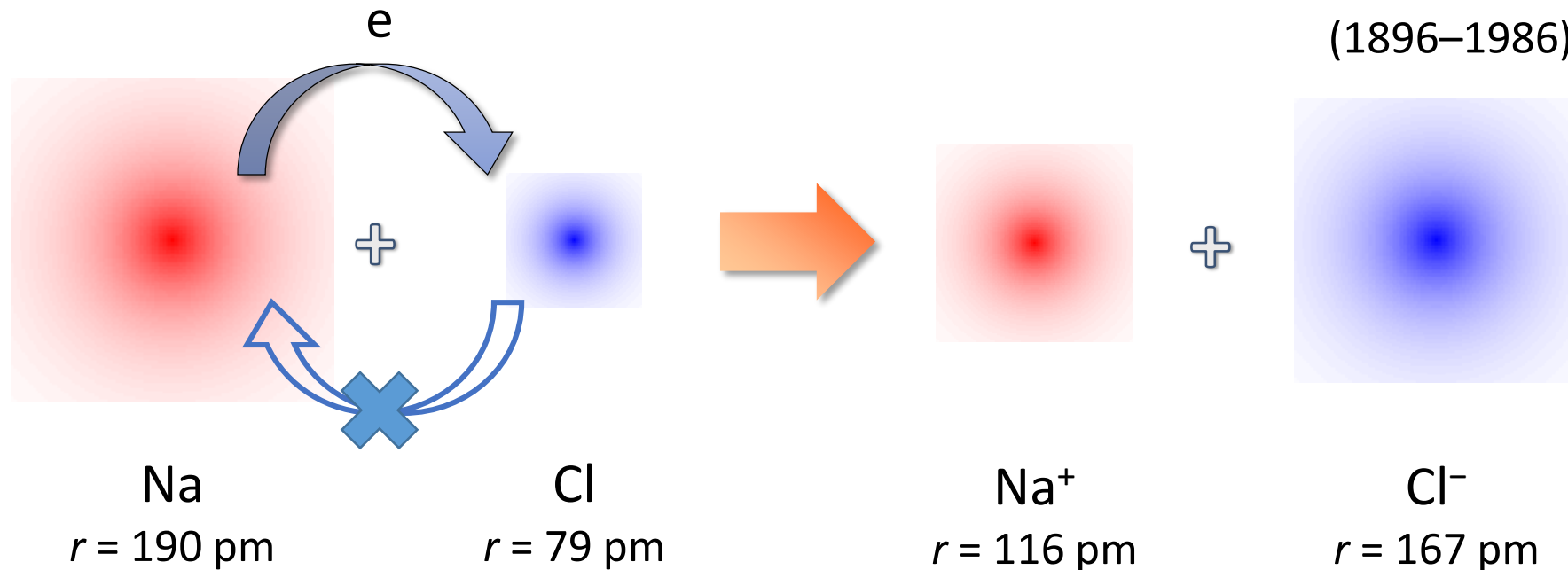


Mulliken's Electronegativity (1934)

电负性

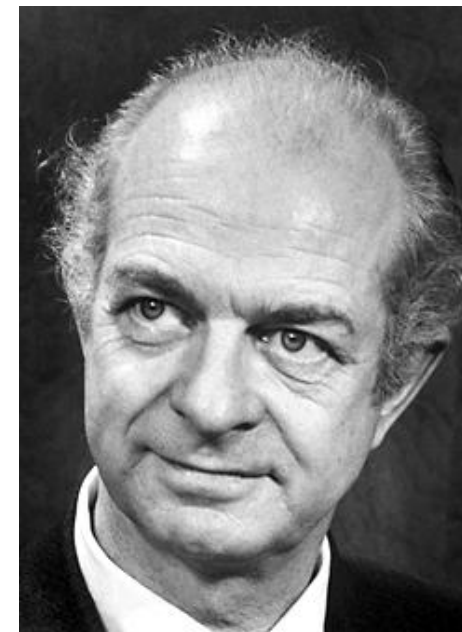
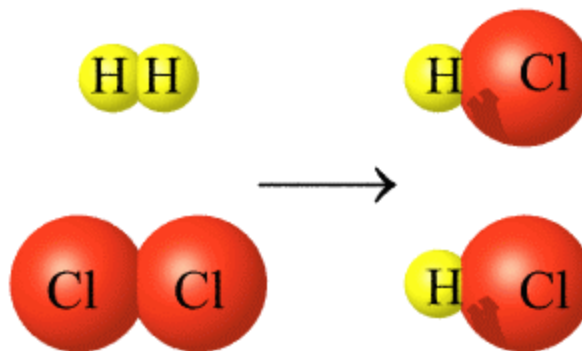
Tendency to attract electrons from a nearby atom.

$$\chi \propto \frac{1}{2}(IE_1 + EA)$$

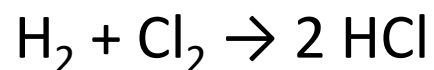


Robert S. Mulliken
(1896–1986)

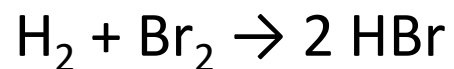
Pauling's Electronegativity (1932)



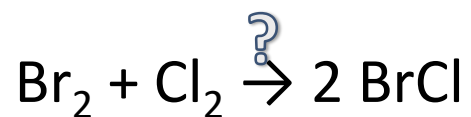
Linus Pauling
(1901–1994)



$$\Delta E = -95 \text{ kJ/mol} = -0.98 \text{ eV/molecule}$$
$$|\chi_{\text{Cl}} - \chi_{\text{H}}| \approx \sqrt{0.98} = 0.99$$

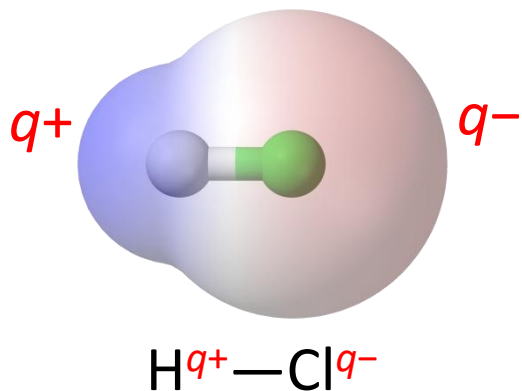


$$\Delta E = -53 \text{ kJ/mol} = -0.55 \text{ eV/molecule}$$
$$|\chi_{\text{Br}} - \chi_{\text{H}}| \approx \sqrt{0.55} = 0.74$$

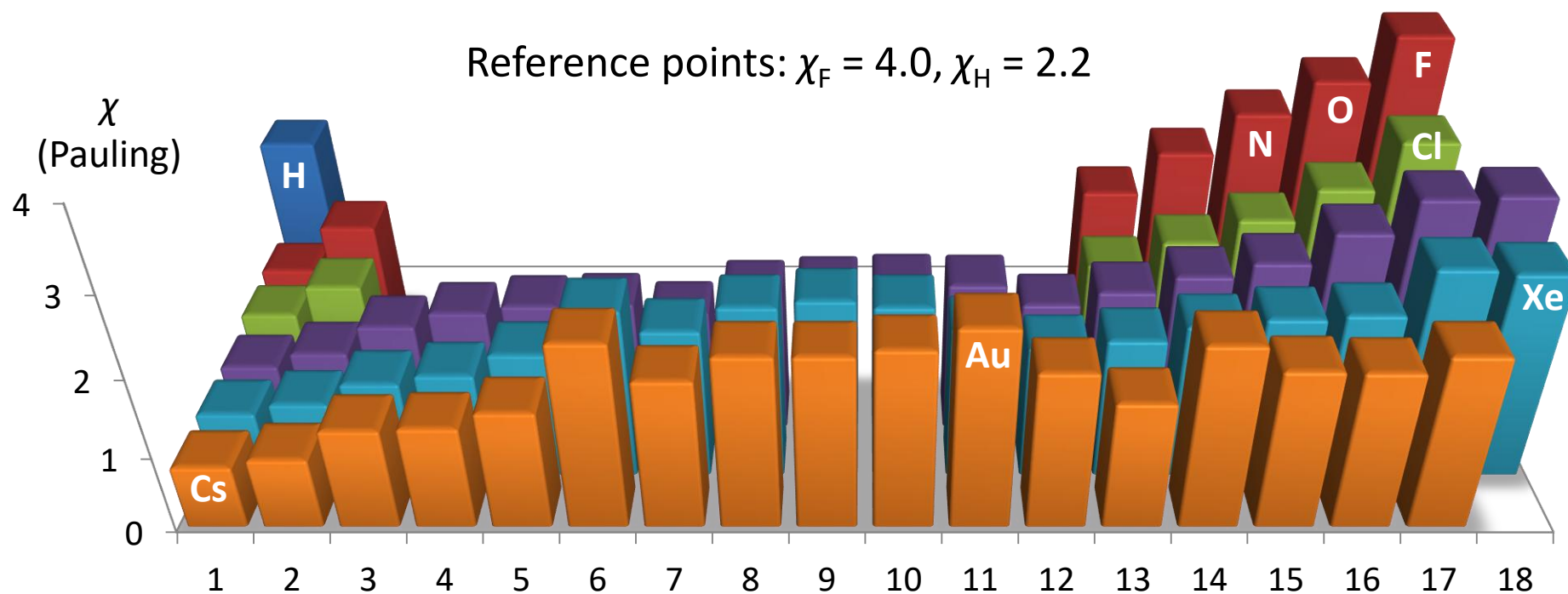


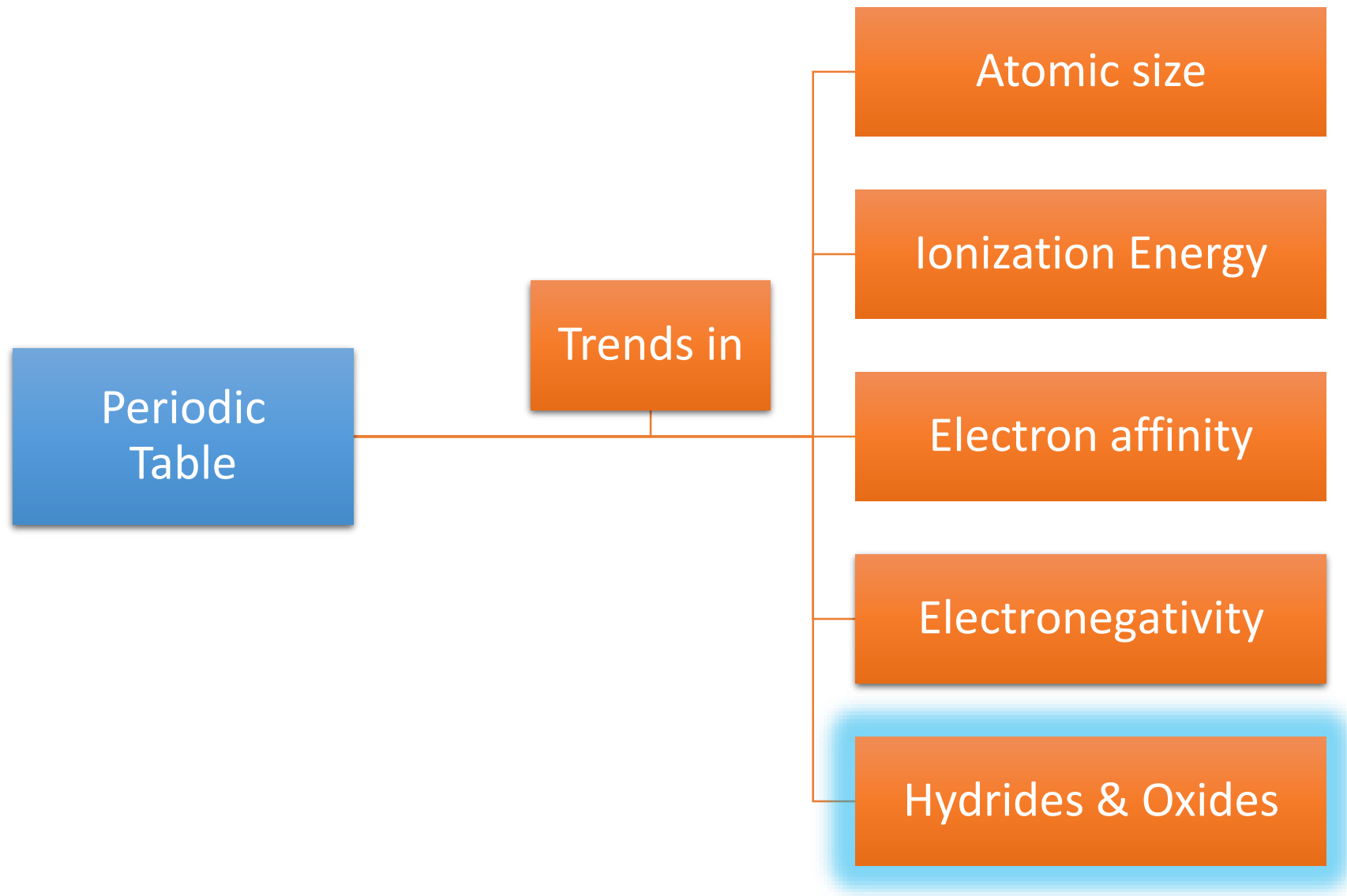
$$\Delta E = -1.0 \text{ kJ/mol} = -0.01 \text{ eV/molecule}$$
$$|\chi_{\text{Cl}} - \chi_{\text{Br}}| \approx \sqrt{0.01} = 0.1$$

Calculating Electronegativities

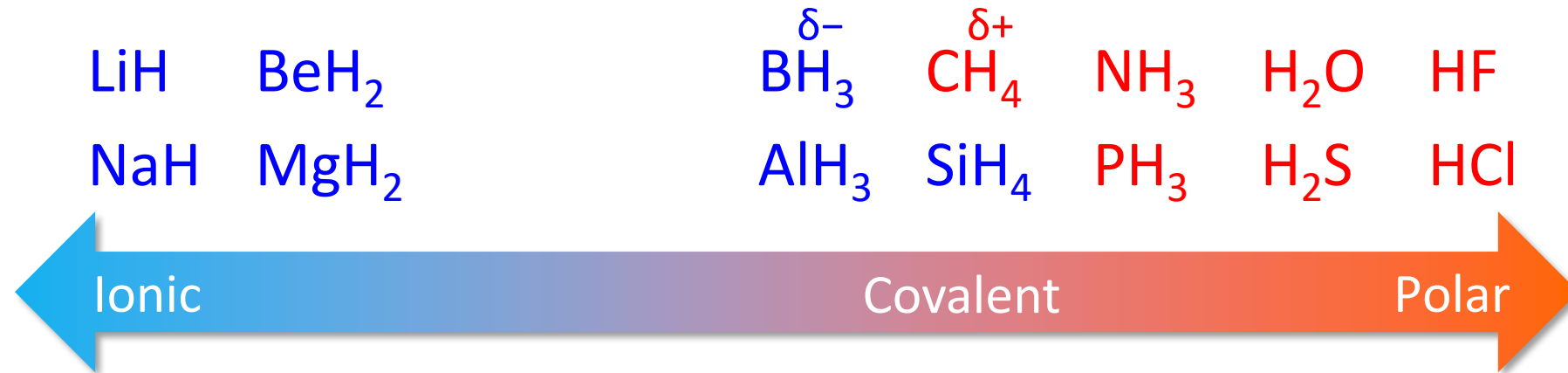


$$(\chi_A - \chi_B)^2 \propto q^2 \propto \Delta E$$





Periodic Trends in Forming Hydrides



H																		He
Li	Be											B	C	N	O	F	Ne	
Na	Mg											Al	Si	P	S	Cl	Ar	
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr	
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe	
Cs	Ba	*	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn	
Fr	Ra	**																

$\chi = 2.20$

$\chi = 2.55$

$$\chi = 2.20$$

$$\chi = 2.55$$

Periodic Trends in Forming Oxides



H																He	
Li	Be											B	C	N	O	F	Ne
Na	Mg											Al	Si	P	S	Cl	Ar
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
Cs	Ba	*	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
Fr	Ra	**															

Acid/Base Properties of Oxides

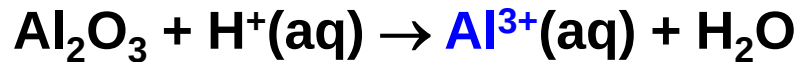
Basic:



Acidic:



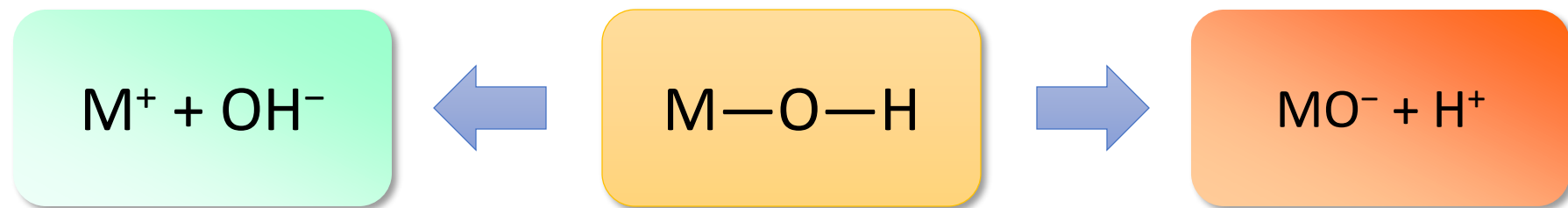
Amphoteric 两性的:



aq: aqueous 水合

Most acidic: HClO_4
Most basic: CsOH

H																	He
Li	Be	<i>Basic</i>										<i>Acidic</i>					
Na	Mg											B	C	N	O	F	Ne
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Al	Si	P	S	Cl	Ar
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	Ga	Ge	As	Se	Br	Kr
Cs	Ba	*	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	In	Sn	Sb	Te	I	Xe
												Tl	Pb	Bi	Po	At	Rn



Summary

- Electronegativity
 - Mulliken: $\chi \propto \frac{1}{2}(\text{IE}_1 + \text{EA})$
 - Pauling: $(\chi_A - \chi_B)^2 \propto q^2 \propto \Delta$
- Bonding
 - Ionic
 - Covalent / polar covalent
- Oxides and hydrides
 - Charge
 - Basic, acid, amphoteric

$$\chi_{\text{F}} = 4.0$$
$$\chi_{\text{H}} = 2.2$$

Consider the elements selenium (Se) and bromine (Br). Which has the higher first ionization energy? Which has the higher electron affinity?

Homework: read “principle of modern chemistry”, chapter 5.4; 5.5